2

CALCULUS

2.1 VECTOR CALCULUS

Calculus refers to the branch of mathematics, which studies the continuous variations of the given function in space and is used to find the rate of change of function in real time. It is used in the development of a mathematical model for getting the optimal solution of motion of astronomical bodies, weather patterns and movement of sound and light. It is classified into two types known as differential and integral calculus. The elements of differential and integral calculus extend to vector fields in a natural way and the branch is known as vector calculus. It deals with the differentiation and integration of vector field. Vector field represents the distribution of a given vector to each point in the subset of space and are often used to model the speed and direction of a moving fluid. Gravitational, magnetic and electric fields are few examples of vector fields as they change their direction and magnitude from point to point.

Before going through the details of vector calculus, a brief description to the function of single variable and multi-variables will be reviewed.

A function of single variable, example, $y = f(x)$ helps in determining the value of y associated with the given x, however a multivariable vector function is designated as $\vec{u} = f(x, y, z)$ along with the direction and notifies about the value of u corresponding to given values of all the variables, i.e. x, y and z. The concept could be easily understood geometrically considering x, y, and z to be the Cartesian coordinates. As an illustration, a function $\mathrm{V}(x, y, z)$ depicts the velocity of air blowing in a room at any point (x, y, z) in the given direction as shown in figure.

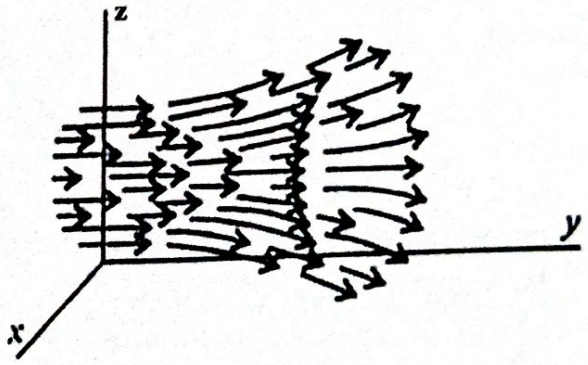


Figure 1: Flow of air in Room. The arrow direction at any point specifies the direction of the vector function

A vector function can be resolved into its components, like any vector as

$$\vec{V}(x,y,z) = V_x(x,y,z)\hat{i} + V_y(x,y,z)\hat{j} + V_z(x,y,z)\hat{k}$$

The three quantities V_x, V_y and V_z, the scalar functions of x, y, and z and are called the components of a vector function . Its value varies in the Euclidean space corresponding to each value of x, y and z. Thus, it is important to analyse these variations to study the behaviour of a vector field in space by calculating the derivatives and integrals of vector fields.

2.2 DERIVATIVE OF A VECTOR FIELD

The derivative of vector-valued functions can also be defined in terms of its component functions and is a straight forward extension of scalar fields. A vector field f is differentiable at $a \in D$, (the domain of the scalar function, comprising the vector field) if each of the component functions f_x, f_y and f_z is differentiable at a. Using the definition of derivative, the differentiation of the function is given as

$$\frac{d\vec{f}}{dr} = \lim_{\Delta r \to a} \frac{\vec{f}(r+\Delta r) - \vec{f}(r)}{\Delta r}$$

that turns out to be simply the derivate of the coefficients of vectors, i.e.

$$\frac{d\vec{f}}{dr} = \frac{\partial f_x}{\partial x}\hat{i} + \frac{\partial f_y}{\partial y}\hat{j} + \frac{\partial f_z}{\partial z}\hat{k} \qquad \ldots(1)$$

Here $\frac{\partial f_x}{\partial x}, \frac{\partial f_y}{\partial y'}$ and $\frac{\partial f_z}{\partial z}$ are the partial derivatives of the components of vectors and are defined as

$$\frac{\partial f_x}{\partial x} = \mathrm{Lt}_{\Delta x \to 0} \frac{f_x(x+\Delta x, y, z) - f_x(x,y,z)}{\Delta x}$$

The derivative of a vector can be interpreted geometrically as the variation in the vector components in a given direction with respect to the independent variables.

The vector-valued functions can be used to represent physical quantities, such as velocity, acceleration, force, momentum, etc. For example, consider an object of constant mass ' m ' has been subjected to some force to move it in space from its initial position (x,y,z) and time ' t ' has been elapsed from some initial time $(t=0)$. Thus, this information can be utilized to define various physical quantities associated with the object as follows:

Position: $\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$

Velocity: $\vec{v}(t) = \vec{r}(t) = \frac{d\vec{r}}{dt} = \frac{dx(t)}{dt}\hat{i} + \frac{dy(t)}{dt}\hat{j} + \frac{dz(t)}{dt}\hat{k}$

Acceleration: $\vec{a}(t) = \vec{v}(t) = \frac{d\vec{v}}{dt} = \frac{dv_x(t)}{dt}\hat{i} + \frac{dv_y(t)}{dt}\hat{j} + \frac{dv_z(t)}{dt}\hat{k},$

$$\vec{\nabla}\phi = \frac{\partial\phi}{\partial x}\hat{i} = \frac{d\phi}{dx}\hat{i}$$

i.e., $$\left|\vec{\nabla}\phi\right| = \frac{d\phi}{dx}$$

in which vector $\vec{\nabla}\phi$ is perpendicular to surface $\phi =$ constant, i.e., is along X-axis and its magnitude $\left(\frac{d\phi}{dx}\right)$ is equal to the maximum rate of change in that direction.

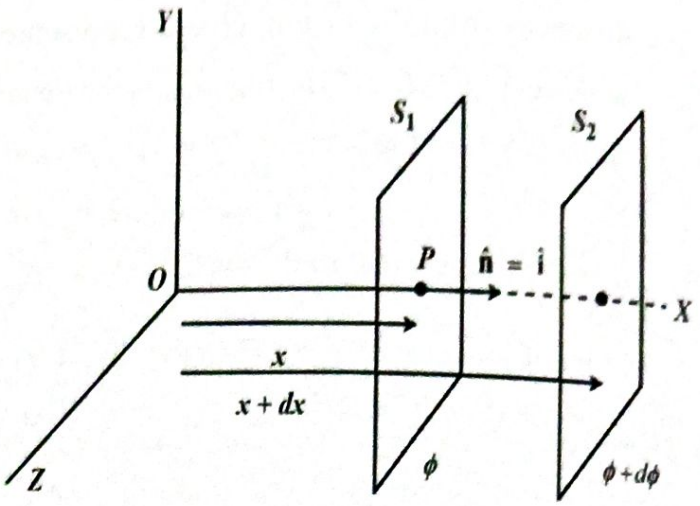


Figure 4

2.5 PHYSICAL SIGNIFICANCE OF GRADIENT

The gradient of a scalar field has a special importance in various fields of physics. For example: if due to a point charge the electric potential at distance r is V, then the equipotential surface will be scalar. The intensity of electric field at a point P situated at a distance r from Q is obtained from electric potential V by the following relation :

$$\vec{E} = -\vec{\nabla}V = -\text{grad}V \qquad \ldots(1)$$

in which potential $V(x,y,z)$ represents a scalar field and the intensity $\vec{E}(x,y,z)$ represents a vector field. If $\hat{r}$ is a unit vector in the direction of $\overrightarrow{OP}$ (Fig. 6), then

$$\vec{E} = -\vec{\nabla}V = -\frac{\partial V}{\partial r}\hat{r} = -\frac{dV}{dr}\hat{r} \qquad \ldots(2)$$

in which $\vec{E}$ is in the direction of unit vector $\hat{r}$ and clearly perpendicular to the equipotential spherical surface. Here, the negative gradient $\left(-\vec{\nabla}V\right)$ of electric potential (scalar field) tells the intensity (vector field) of electric potential. The intensity of field at any point P is a vector which is in the direction of the greatest rate of decrease of potential, i.e., in the direction of unit vector $\hat{r}$ and whose magnitude is equal to the greatest rate of change $\frac{dV}{dr}$ of potential.

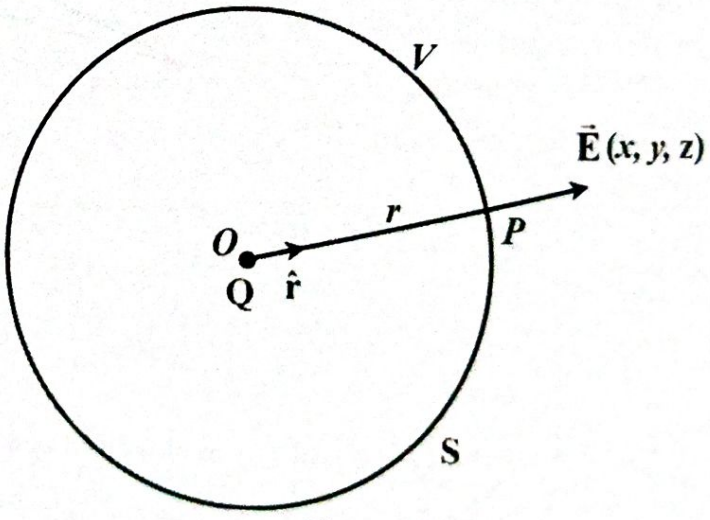


Figure 5

Similarly, the intensity of any gravitational field $\vec{E}$ is related with gravitational potential V by $\vec{E} = -\vec{\nabla}V$. Now, if a charge q rests at a point P then its potential energy

will be U (which is equal to qV) and the force $\vec{F}$ on that charge will be the negative gradient of potential energy U, i.e..

$$\vec{F} = -\overrightarrow{VU} = -\text{grad}U = -\frac{\partial U}{\partial r}\hat{r} = -\frac{dU}{dr}\hat{r} \qquad \ldots(3)$$

2.6 DIVERGENCE OF A VECTOR FIELD

If $\vec{A}(x,y,z)$ is a vector field, the scalar product of the vector operator $\vec{\nabla}$ and $\vec{A}$ is a scalar and is called the divergence of $\vec{A}$, i.e.

$$\text{div}\vec{A} = \vec{\nabla}\cdot\vec{A} = \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)\left(\hat{i}A_x + \hat{j}A_y + \hat{k}A_z\right)$$

or

$$\text{div}\vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = \vec{\nabla}\cdot\vec{A} \qquad \ldots(1)$$

This quantity has important applications in hydrodynamics and electricity.

We shall show in the following discussion that the quantity $\vec{\nabla}\cdot\vec{A}$ at a point is the amount of flux per unit volume diverging from that point. Hence, the name divergence originated from the interpretation of $\vec{\nabla}\cdot\vec{A}$.

2.7 PHYSICAL INTERPRETATION AND SIGNIFICANCE OF DIV $\vec{A}$

To understand the meaning and physical importance of the divergence of a vector field, we take an example of flow of a liquid. In general, this liquid is compressible. Here, the velocity of liquid $(\vec{v})$ represents the vector field $A(x,y,z)$. Let $\vec{A}(x,y,z)$ be the vector field at the mid-point of a small rectangular parallelepiped $ABCDEFGH$ (whose volume is $dV = dxdydz$). The axial components of the vector $\vec{A}$ and A_x, A_y and A_z and in general, the value of this vector at different point of the region is different. We may think $\vec{A}$ as the velocity of a liquid.

On the face $ABCD$ the value of the X-component of the vector $\vec{A}$ is

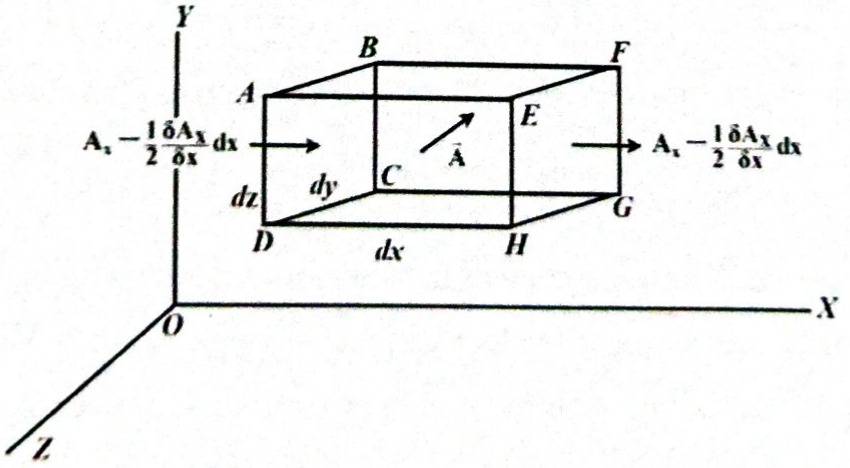


Figure 6

As the face is infinitely small, this value can be taken as uniform over the face $ABCD$. Similarly for the face $EFGH$, the $X-$component is

$$A_1+\frac{1}{2}\frac{dd_1}{dy}dx$$

The flux through a face is defined as the product of the area of the face and the normal component of the vector upon it.

$\therefore$ Flux entering the face $ABCD=\left(A_1-\frac{1}{2}\frac{\partial d_1}{\partial x}dx\right)dx$

The vector $\vec{A}$ is representing the velocity of liquid. Therefore, the volume of the liquid flowing (entering) per second in the face $ABCD$) perpendicular to X -axis will be given as above.

$$\text{The flux leaving the face } EFGH=\left(A_1+\frac{1}{2}\frac{\partial A_1}{\partial x}dx\right)dydz$$

Hence, the excess of flux leaving the parallelopiped over that entering it in the X - direction is

$$\left(A_x+\frac{1}{2}\frac{\partial A_1}{\partial x}dx\right)dydz-\left(A_x-\frac{1}{2}\frac{\partial A_x}{\partial x}dx\right)dydz=\frac{\partial A_x}{\partial x}dxdydz$$

In the hydromechanical case, this represents the net volume of the fluid passing per second in the Y -direction. Similarly, contributions parallel to Y and Z directions are $\frac{\partial A_y}{\partial y}dxdydz$ and $\frac{\partial t_z}{\partial z}dxdydz$

Hence the net flux diverging from the element is

$$\cdot\ d\phi=\left(\frac{\partial A_x}{\partial x}+\frac{\partial A_y}{\partial y}+\frac{\partial A_z}{\partial z}\right)dxdydz$$

$$=\left(\vec{\nabla}\cdot\vec{A}\right)dxdydz=\left(\text{div}\vec{A}\right)dxdydz=\left(\text{div}\vec{A}\right)dV$$

This quantity denotes the net volume diverging per second for volume dV. Hence

$$\frac{d\phi}{dV}=\text{div}\vec{A}=\vec{\nabla}\cdot\vec{A} \qquad \ldots(2)$$

Thus, $\vec{\nabla}\cdot\vec{A}$ (or $\text{div}\vec{A}$) at a point gives the amount of flux diverging per unit volume from that point. If $\vec{A}$ is denoting the velocity of the moving flux, then $\vec{\nabla}\cdot\vec{A}$ tells the rate at which the fluid is diverging form the point per unit volume. The Vector function $\vec{A}$ may be the velocity of a moving gas also. So, we may use the word fluid also in place of liquid.

2.8 PHYSICAL SIGNIFICANCE OF DIVERGENT

If the flux, entering any element of fluid space is exactly balanced by that leaving it, the quantity $\text{div}\vec{A} = 0$. Then, there is no source or sink, nor its density is changing, i.e., the fluid is incompressible. If the fluxes entering an leaving an element are equal, the lines of flow of the vector $\vec{A}$ should form either closed curves (e.g., in case of magnetic field of a current) or extend to infinity. A vector which satisfies this condition is called solenoidal.

For example, one of the Maxwell's electromagnetic field equations is

$$\text{div}\vec{B} = 0 \text{ or } \vec{\nabla}\cdot\vec{B} = 0$$

in which magnetic induction $\vec{B}$ is a solenoidal vector.

2.9 VECTOR INTEGRATION

The integrals, which we generally come across in Vector Algebra are the line integral, the surface integral and the **volume integral.**

(1). Line integral: Let PQ be any curve, drawn in a vector field and $d\vec{l}$ an element of length along this curve at any point B. Let $\vec{A}$ represent the vector at B in a direction making an angle θ with $d\vec{l}$. In general if $\vec{A}$ varies in magnitude and direction from point to point along the curve, the integral

$$\int_P^Q \vec{A}\cdot d\vec{l} = \int_P^Q A\cos\theta\, dl \qquad \ldots(1)$$

is defined as the *line integral* of $\vec{A}$ along the curve PQ.

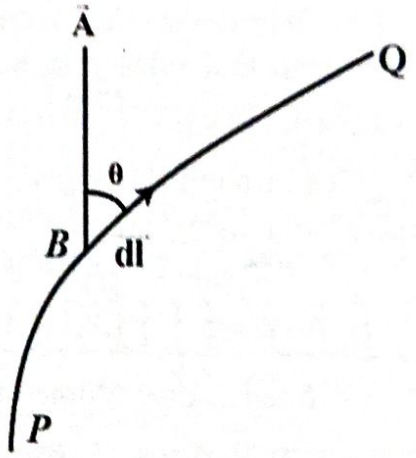


Figure 7

In terms of Cartesian components,

$$\int_P^Q \vec{A}\cdot d\vec{l} = \int_P^Q \left(A_x dx + A_y dy + A_z dz\right) \qquad \ldots(2)$$

If a force $\vec{F}$ is acting on a moving particle then the line integral $\int \vec{F}\cdot d\vec{l}$ over the path described by the particle is the work done by the force, i.e.,

$$W = \int_P^Q \vec{F}\cdot d\vec{l} \qquad \ldots(3)$$

Similarly, if a unit charge be taken from P to Q in any electric field $\vec{E}$, then line integral $\int \vec{E}\cdot d\vec{l}$ represents the potential difference between these two points, i.e.,

$$\Delta V = -\int_P^Q \vec{E}\cdot d\vec{l} \qquad \ldots(4)$$

We shall study later that if the vector field is conservative, then the value of the line integral along the closed path is zero, i.e., in conservative field.

$$\oint \vec{A}\cdot d\vec{l} = 0 \qquad \ldots(5)$$

(2). Surface integral: Consider any element of infinitesimal area $d\vec{s}$ upon a surface S in a vector field. If $\vec{A}$ be the value of the vector at the middle point of the element, then the integral over the surface

$$\iint_S \vec{A}\cdot d\vec{s} = \iint_S A\cos\theta\, ds \qquad \ldots(6)$$

is defined as the surface integral or total flux of vector $\vec{A}$ through the whole surface S.

In Cartesian components,

$$\iint_S \vec{A}\cdot d\vec{s} = \iint_S \left(A_x ds_x + A_y ds_y + A_z ds_z\right) \ldots(7)$$

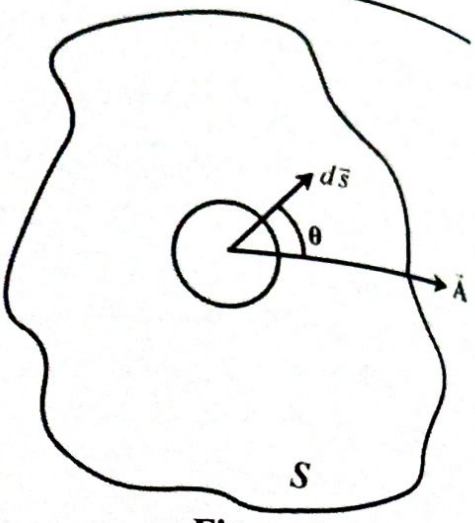


Figure 8

If $\vec{V}$ denotes the vector velocity of a moving fluid, in which a fixed surface S is drawn, then $\iint_S \vec{V}\cdot d\vec{s}$ will represent the rate of flow of liquid through the surface S, i.e., represents the volume of the liquid through that surface in unit time. In electric field $\vec{E}$, integral $\iint_S \vec{E}\cdot d\vec{s}$ represents the flux going through surface S.

(3). Volume integral: If we consider a closed surface in space enclosing a volume V, then

$$\iiint_V \vec{A}\,d\vec{V} = \int_x\int_y\int_z \left(A_x\hat{i} + A_y\hat{j} + A_z\hat{k}\right)dx\,dy\,dz$$

is defined as the volume integral.

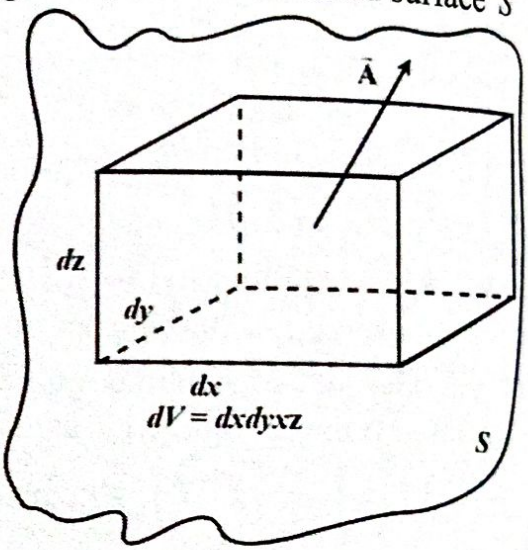


Figure 9

2.10 CURL

If $\vec{A}(x,y,z)$ is a vector field, then cross product of the operator $\vec{\nabla}$ and $\vec{A}$ is a vector and is called curl or rotation of $\vec{A}$. Thus,

$$\text{curl}\vec{A} = \vec{\nabla}\times\vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

$$= \hat{i}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \hat{j}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + \hat{k}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)$$

The concept of curl is very useful in Vector Calculus and has significant and wide applications in hydrodynamics and electro-magnetism.

The curve of vector region $\vec{A}$ at any point is related with the line integral $\left(\oint \vec{A}\cdot d\vec{l}\right)$ taken along the boundary of an infinitesimally small plane area around that point and curl $\vec{A}$ is defined as follows:

The curl or rotation at any point in vector region $\vec{A}$, i.e., curl $\vec{A}$ is a vector quantity whose magnitude is equal to the maximum magnitude of line integral $\oint \vec{A}\cdot d\vec{l}$ for unit area along the boundary C of infinitesimally small area around that point and the direction is perpendicular to that infinitesimally small area.

2.11 PHYSICAL INTERPRETATION AND SIGNIFICANCE OF CURL $\vec{A}$

To understand the physical importance of curl of any vector region, one consider the circulation of a fluid along an infinitesimally small area in XY plane. In hydrodynamics, the line integral of any vector region $\vec{A}$ (Let the velocity of liquid $\vec{v}$ as in whirlpools and eddies) along a closed curve is called the circulation of $\vec{A}$ along that path. For the sake of convenience, we take this boundary in the form of an infinitesimally small rectangle. In Fig 10 , it is at rectangle $PQRS$ where sides δx and δy lie in XY plane.

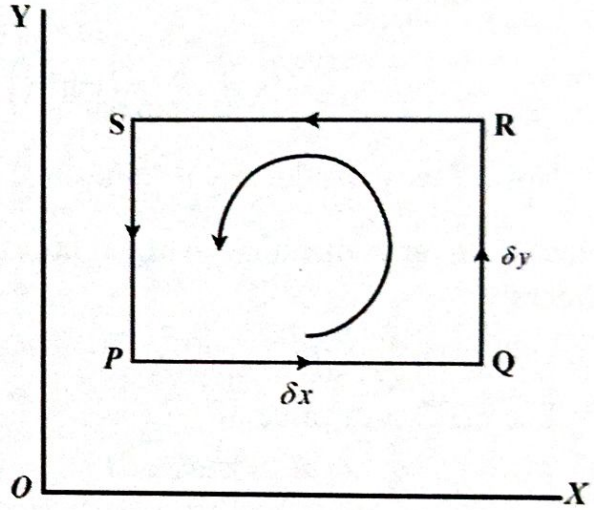


Figure 10

Let at the point P, x and y components of $\vec{A}$ are A_x and A_y respectively.

The average value of the x-components of $\vec{A}$ over the path PQ can be taken

$$A_x + \frac{\partial A_x}{\partial x}\cdot\frac{\delta x}{2}$$

and over the path RS as

$$A_x + \frac{\partial A_x}{\partial x}\cdot\frac{\delta x}{2} + \frac{\partial A_x}{\partial y}\cdot\delta y$$

$$\therefore \quad \int_P^Q \vec{A}\cdot d\vec{l} + \int_R^S \vec{A}\cdot d\vec{l} = \left(A_x + \frac{\partial A_x}{\partial x}\cdot\frac{\delta x}{2}\right)\delta x - \left(A_x + \frac{\partial A_x}{\partial x}\cdot\frac{\delta x}{2} + \frac{\partial A_x}{\partial y}\delta y\right)\delta x$$

$$= -\frac{\partial A_x}{\partial y}\cdot\delta x\delta y \qquad \ldots(2)$$

because the direction of the path RS is towards the negative direction of X.

Similarly. $\int_Q^R \vec{A}\cdot d\vec{l} + \int_S^P \vec{A}\cdot d\vec{l} = \frac{\partial A_y}{\partial x}\delta x\delta y$ $\ldots(3)$

By adding equations (2) and (3), we get the line integral

$$\oint_{PQRS} \vec{A} \cdot d\vec{l} = \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \delta x \delta y$$

But $\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$ is the z-component of curl $\vec{A}$ or $\nabla \times \vec{A}$ and the infinitesimal area $\delta x \delta y$ may be regarded as the magnitude of the vector area $\delta s_z \hat{k}$. Thus,

$$\oint_{PQRS} \vec{A} \cdot d\vec{l} = (\text{curl}\vec{A})_z \delta s_z = \left(\text{curl}\vec{A}\right) \cdot \delta s_z \hat{k}$$

This equation is correct for any infinitesimal area of magnitude δs_z at P, having boundary C, i.e.,

$$\oint_C \vec{A} \cdot d\vec{l} = \left(\text{curl}\vec{A}\right) \cdot \delta s_z \hat{k}$$

Now, if we consider any infinitesimal plane area $\delta\vec{s} = \delta s_x \hat{i} + \delta s_y \hat{j} + \delta s_z \hat{k}$ at P oriented in any direction with a boundary C, then evidently

$$\oint_C \vec{A} \cdot d\vec{l} = \text{curl}\vec{A} \cdot \vec{s} \qquad \ldots(4a)$$

The maximum value of

$$\oint_C \vec{A} \cdot d\vec{l} = \left|\text{curl}\vec{A}\right| \delta s \qquad \ldots(4b)$$

Thus, the magnitude of curl $\vec{A}$ at a point may be defined as the maximum value of the line integral of $\vec{A}$ along a boundary $C\left(\oint \vec{A} \cdot d\vec{l}\right)$ per unit area; this boundary C is that of an infinitesimal plane area (δs) at the point under consideration.

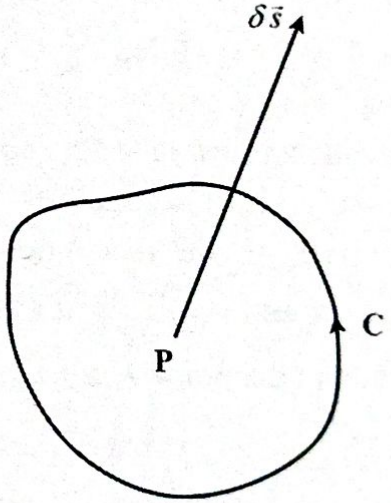


Figure 11

2.12 LAPLACIAN OPERATOR

The operator $\overrightarrow{\nabla^2} = \left(\vec{\nabla} \cdot \vec{\nabla}\right)$ is known as Laplacian operator. It is defined as

$$\nabla^2 = \vec{\nabla} \cdot \vec{\nabla} = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

It is a scalar operator. It can be operated on scalar quantity and then it gives a scalar function (quantity).

2.13 GRADIENT THEOREM

According to gradient theorem, a line integral through a gradient field along a curve is equal to the scalar field at the end points of the curve. This can be written as

$$\int_P^Q \vec{\nabla}\phi \cdot \overrightarrow{dr} = \phi(Q) - \phi(P)$$

Proof : Let $\phi(x, y, z)$ be a continuously differentiable scalar field of position in a region and C_1, be curve which starts at P and ends at Q. By taking the gradient of ϕ, we obtain a vector field $\vec{A}$, given by $\vec{A} = \vec{\nabla}\phi$. Hence the line integral of the vector field $\vec{A}$ along the curve C_1, from point P to Q is given by

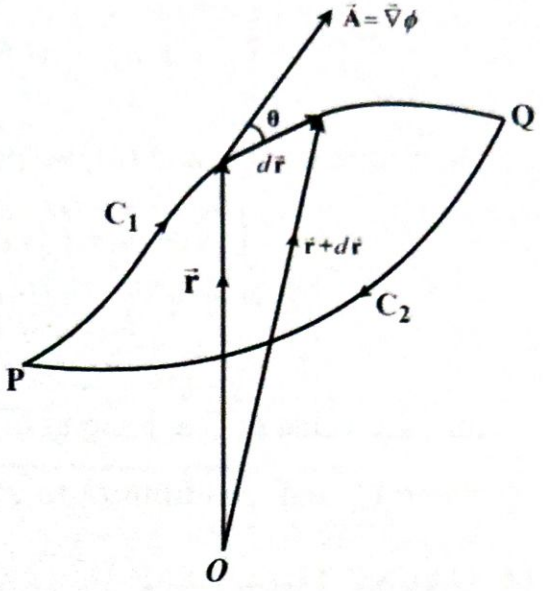


Figure 12

$$\int_P^Q \vec{A} \cdot d\vec{r} = \int_P^Q \vec{\nabla} \cdot d\vec{r}$$

where $d\vec{r}$ is infinitesimal small line element along the curve.

But $$\vec{\nabla}\phi = \frac{\partial \phi}{\partial x}\hat{i} + \frac{\partial \phi}{\partial y}\hat{j} + \frac{\partial \phi}{\partial z}\hat{k}$$

and $$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$$

Therefore $$\int_P^Q{}_{C_1} \vec{\nabla}\phi \cdot d\vec{r} = \int_P^Q{}_{C_1} \left(\frac{\partial \phi}{\partial x}\hat{i} + \frac{\partial \phi}{\partial y}\hat{j} + \frac{\partial \phi}{\partial z}\hat{k}\right) \cdot \left(dx\hat{i} + dy\hat{j} + dz\hat{k}\right)$$

$$= \int_P^Q{}_{C_1} \left(\frac{\partial \phi}{\partial x}dx + \frac{\partial \phi}{\partial y}dy + \frac{\partial \phi}{\partial z}dz\right)$$

$$= \int_P^Q{}_{C_1} d\phi = \phi(Q) - \phi(P)$$

$$\int_P^Q{}_{C_1} \vec{\nabla}\phi \cdot d\vec{r} = \phi(Q) - \phi(P) \qquad \ldots(1)$$

Hence the gradient theorem is proved.

Next, we take the line integral of $\vec{\nabla}\phi$ along curve C_2 from the point P to Q as

$$\int_P^Q{}_{C_1} \vec{\nabla}\phi \cdot d\vec{r} = \phi(Q) - \phi(P) \qquad \ldots(2)$$

From equation. (1) and (2), we find that the value of the line integral along curve C_1 and curve C_2 is the same. This means that the value of line integral of $\vec{\nabla}\phi$ taken between any two points in the scalar field is **independent of the path.**

If we take the line integral of the vector field $\vec{\nabla}\phi$ along the curve C_2 from point Q to P, then

$$\int_P^Q{}_{C_1} \vec{\nabla}\phi \cdot d\vec{r} = \phi(P) - \phi(Q) \quad \ldots(3)$$

By adding equation. (1) and (3), we get

$$\int_P^Q \vec{\nabla}\phi \cdot d\vec{r} + \int_Q^P \vec{\nabla}\phi \cdot d\vec{r} = 0$$

Or
$$\oint \vec{\nabla}\phi \cdot d\vec{r} = 0 \quad \ldots(4)$$

Thus, the value of line integral of $\vec{\nabla}\phi$ or grad ϕ along a closed path (i.e., from P to Q along C_1 and then from Q to P along C_2) is zero.

2.14 GAUSS' THEOREM

Statement : ***According to this theorem, the volume integral of the divergence of a vector field $\vec{A}$ taken over any volume V is equal to the surface integral of $\vec{A}$ taken over the closed surface surrounding the volume. That is***

$$\iiint_V \left(\vec{\nabla}\cdot\vec{A}\right) dV = \iint_S \vec{A}\cdot\overrightarrow{d\,s} \quad \ldots(1)$$

Proof: In Cartesian coordinates,

$$\vec{\nabla}\cdot\vec{A} = \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)\cdot\left(A_x\hat{i} + A_y\hat{j} + A_z\hat{k}\right)$$

$$= \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

And
$$dV = dxdydz$$

$$\vec{A}\cdot d\vec{s} = \left(\left(A_x\hat{i} + A_y\hat{j} + A_z\hat{k}\right)\right)\cdot\left(ds_x\hat{i} + ds_y\hat{j} + ds_z\hat{k}\right)$$

$$= A_x ds_x + A_y ds_y + A_z ds_z$$

$$= A_x dydz + A_y dxdz + A_z dxdz.$$

Hence, Gauss' theorem in Cartesian coordinates is

$$\iiint_V \left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}\right) dx\,dy\,dz = \iint_S \left(A_x ds_x + A_y ds_y + A_z ds_z\right) \ldots(2)$$

Let us consider the first part of the left-hand side integral. As shown in Fig. 16 integrate it with respect to x along a strip of cross-section $dydz$ extending from $P_1(x_1, y, z)$ to $P_2(x_2, y, z)$, i.e.

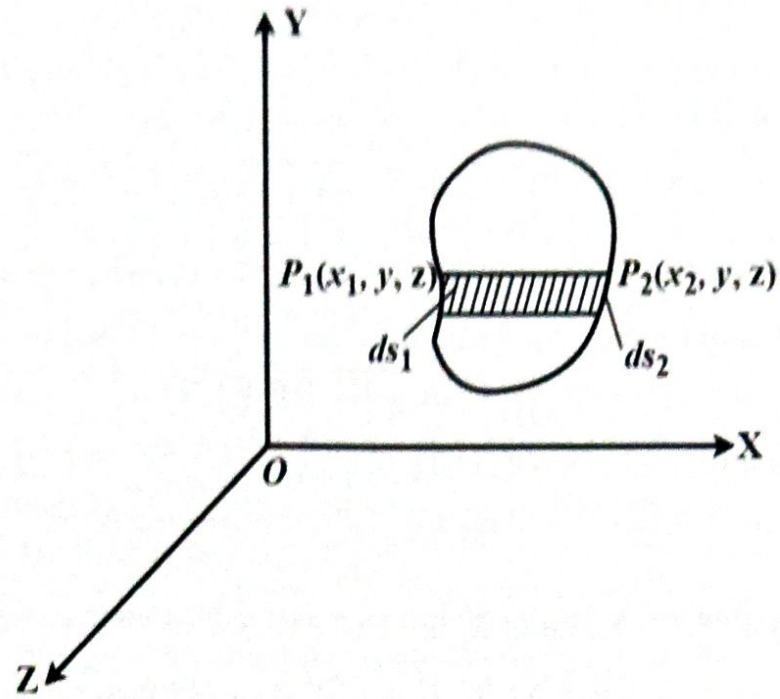


Figure 13

$$\iiint_V \frac{\partial A_x}{\partial x}\,dx\,dy\,dz = \iint_S [A_x]_A^{\beta_2}\,dy\,dz$$

$$= \iint_S \left[A_x(P_2) - A_x(P_1)\right] dydz$$

$$= \iint_S \left[A_x(x_2, y, z) - A_x(x_1, y, z)\right] dy\,dz \qquad \ldots(3)$$

Evidently, at P_1, $A_x dydz = -A_x ds_x$ and at P_2, $A_x dydz = A_x ds_x$

(Since at P_1 and P_2, the X-components of area have opposite directions.)

$\therefore$ $$\iiint_V \frac{\partial A_x}{\partial x}\,dx\,dy\,dz = \iiint_S A_x ds_x \qquad \ldots(4a)$$

Similarly, $$\iiint_V \frac{\partial A_y}{\partial y}\,dxdydz = \iint_S A_y ds_y \qquad \ldots(4b)$$

and $$\iiint_V \frac{\partial A_y}{\partial z}\,dxdydz = \iiint_S A_z ds_z \qquad \ldots(4c)$$

Hence, adding equations (4a), (4b) and (4c), we have

$$\iiint_V \left(\frac{\partial A_x}{\partial x} + \frac{\partial y_y}{\partial y} + \frac{\partial A_z}{\partial z}\right) dx\,dy\,dz = \iint_S \left(A_x ds_x + A_y ds_y + A_z ds_z\right)$$

$$\iiint_V \left(\vec{\nabla} \cdot \vec{A}\right) dV = \iint_S \vec{A} \cdot d\vec{s} \qquad \ldots(5)$$

or $$\iiint_V \left(\text{div}\vec{A}\right) dV = \iint_S \vec{A} \cdot d\vec{s}$$

This proves the Gauss' theorem of divergence.

2.15 APPLICATION OF GAUSS' THEOREM

Applying this theorem for electrostatic field one get an important law of electrostatics. For distributed charge, [1]Gauss' law is

$$\iint_S \vec{E}\cdot\overrightarrow{ds} = \frac{1}{\varepsilon_0}\iiint_V \rho dV \qquad \ldots(6)$$

where ρ is the surface enclosing volume V and ρ is charge density and ε_0 is the permittivity of vacuum (or air). Applying divergence theorem in this equation,

$$\iint_S \vec{E}\cdot\overrightarrow{ds} = \iiint_V \left(div\vec{E}\right)dV \qquad \ldots(7)$$

Comparing equation (6) and (7), we get

$$\text{div}\vec{E} = \frac{\rho}{\varepsilon_0}$$

Equation (8) is the differential form of Gauss' law in electrostatics. According to this, the divergence of electric field $\vec{E}$ at any point in $\frac{1}{\varepsilon_0}$ times the charge in per unit volume near to that point.

Among Maxwell's electromagnetics field equations, one equation is the relation (8) also.

2.16 GREEN'S THEOREM

Statement: ***We can make some important transformations by the use of Gauss's theorem. Let us consider that the vector field*** $\vec{A}$ ***be the product of scalar function*** ϕ_1 ***and the gradient of another scalar function*** ϕ_2***, i.e.,***

$$\vec{A} = \phi_1\vec{\nabla}\phi_2 = \hat{i}\,\phi_1\frac{\partial\phi_2}{\partial x} + \hat{j}\,\phi_1\frac{\partial\phi_2}{\partial y} + \hat{k}\,\phi_1\frac{\partial\phi_2}{\partial z} \qquad \ldots(1)$$

$$= \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$= \frac{\partial}{\partial x}\left(\phi_1\frac{\partial\phi_2}{\partial x}\right) + \frac{\partial}{\partial y}\left(\phi_1\frac{\partial\phi_2}{\partial y}\right) + \frac{\partial}{\partial z}\left(\phi_1\frac{\partial\phi_2}{\partial z}\right)$$

$$= \phi_1\left(\frac{\partial^2\phi_2}{\partial x^2} + \frac{\partial^2\phi_2}{\partial y^2} + \frac{\partial^2\phi_2}{\partial z^2}\right) + \frac{\partial\phi_1}{\partial x}\cdot\frac{\partial\phi_2}{\partial x} + \frac{\partial\phi_1}{\partial y}\frac{\partial\phi_2}{\partial y} + \frac{\partial\phi_1}{\partial z}\frac{\partial\phi_2}{\partial z}$$

---

[1] *The reader should not confuse that we have written for

$$\iiint [A_x(P_2) - A_x(P_1)]\,dydz = \iint (A_x ds_x + A_x ds_x) \text{ only } \iint A_x ds_x$$

because if we take all the strips then each integral $\iint A_x ds_x$ will account for the half surface integral. Hence $\iint A_x ds_x$ accounts for the total surface integral for X--component.

$$= \phi_1 \nabla^2 \phi_2 + \nabla\phi_1 \cdot \nabla\phi_2 \qquad \ldots(2)$$

Where $\nabla^2 = \dfrac{\partial^2}{\partial x^2} + \dfrac{\partial^2}{\partial y^2} + \dfrac{\partial^2}{\partial z^2}$

Now, by Gauss' theorem,

$$\iiint_V \left(\vec{\nabla} \cdot \vec{A}\right) dV = \iint_S \vec{A} \cdot d\vec{s} \qquad \ldots(3)$$

Putting the value of $\left(\vec{\nabla} \cdot \vec{A}\right)$ in LHS of (3), we get.

Substituting for $\vec{\nabla} \cdot \vec{A}$ in eq. (3), we get

$$\iiint_V \left(\phi_1 \nabla^2 \phi_2 + \nabla\phi_1 \cdot \nabla\phi_2\right) dV = \iint_S (\phi_1 \nabla\phi_2) \cdot d\vec{s}$$

We call the transformation represented by equation (4) as the first form of Green's theorem.

If we mutually interchange the scalar functions ϕ_1 and ϕ_2 in equation (4), we obtain

$$\iiint_V \left(\phi_2 \nabla^2 \phi_1 + \nabla\phi_2 \cdot \nabla\phi_1\right) dV = \iint_S (\phi_2 \nabla\phi_1) \cdot d\vec{s}$$

Subtracting equation (5) from equation (4), we get

$$\iiint_V \left(\phi_1 \nabla^2 \phi_2 - \phi_2 \nabla^2 \phi_1\right) dV = \iint_S (\phi_1 \nabla\phi_2 - \phi_2 \nabla\phi_1) \cdot d\vec{s}$$

Equation (6) represents the second form of Green's theorem. These transformation equations (4) and (6), obtained by the use of Gauss' theorem, find important applications in the field of electro-dynamics and hydro-dynamics.

2.17 STROKE'S THEOREM

Statement: ***This theorem states that the line integral of a vector $\vec{A}$ around any closed curve C is equal to the surface integral of curl $\vec{A}$ taken over any surface S of which the curve is a boundary edge, i.e.,***

$$\oint_C \vec{A} \cdot d\vec{l} = \iint_S \text{curl}\vec{A} \cdot d\vec{s} = \iint_S \left(\vec{\nabla} \times \vec{A}\right) \vec{s} \qquad \ldots(1)$$

Proof: If a vector $\vec{A}$ be a function of position, then its line integral along a closed curve C is given by

$$L = \oint_C \vec{A} \cdot d\vec{l} \qquad \ldots(2)$$

where $d\vec{l}$ is the small element of the path. Now, divide the enclosed area into two parts by a line pq so as to form two closed curves C_1 and C_2. If L_1 and L_2 are the line integrals for the two paths C_1 and C_2, then their sum $(L_1 + L_2)$ will be equal to L for the curve C. The reason lies in the fact that the part pq is traversed from p to q for the curve C_1 and from q to p for the path C_2 so that its contribution is cancelled. Thus, if we divide the area enclosed by the curve C into a number of smaller areas $\delta s_1, \delta s_2, \ldots$ bounded by curves $C_1, C_2, C_3, \ldots, C_n \ldots$, then

$$\oint_C \vec{A} \cdot d\vec{l} = \sum \oint_{cn} \vec{A} \cdot d\vec{l} \qquad \ldots(3)$$

The curve C may be divided in such a way that the small areas $\delta s_1, \delta s_2$ may lie in different planes.

But from the definition of curl $\vec{A}$ at a point inside δs_n, we have

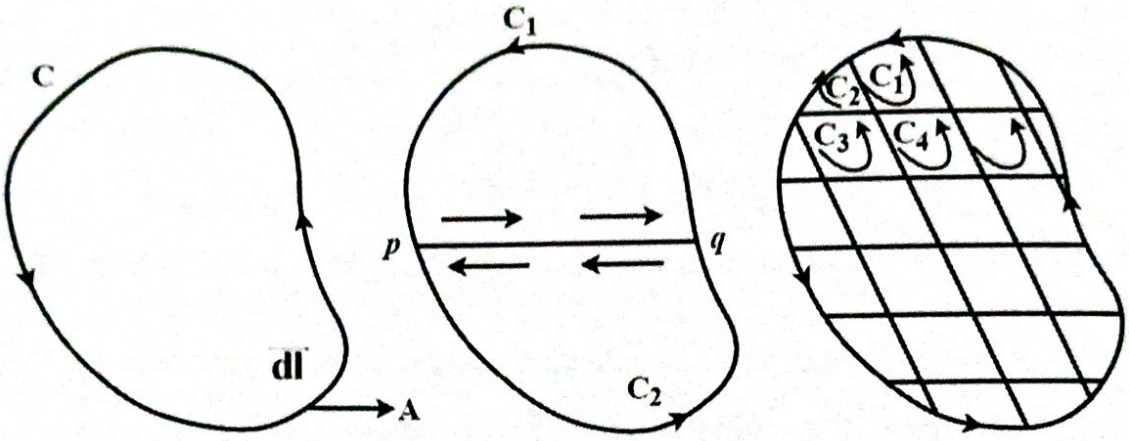


Figure 14

$$\text{curl}\vec{A} \cdot \delta s_n = \oint_{cn} \vec{A} \cdot d\vec{l} \qquad \ldots(4)$$

Thus,
$$\oint_C \vec{A} \cdot d\vec{l} = \sum \oint_{cn} \vec{A} \cdot d\vec{l}$$
$$= \sum \text{curl}\vec{A} \cdot \delta s_n$$

i.e.,
$$\oint_C \vec{A} \cdot d\vec{l} = \int_S \text{curl}\vec{A} \cdot d\vec{s} \qquad \ldots(5)$$

This is Stroke's theorem. It should be noted that the boundary curve or the enclosed surface need not to be plane one.

If the surface to which Stroke's theorem is applied is a closed surface, the length of the boundary edges is zero and then

$$\iint \text{curl}\vec{A} \cdot d\vec{s} = 0 \qquad \ldots(6)$$

In case, curl $\vec{A}$ or $\vec{\nabla} \times \vec{A} = 0$ everywhere, then according to Stroke's theorem the line integral of $\dot{A}$ around any closed curve vanishes. This means that $\dot{A}$ should be the *gradient of a scalar function* ϕ, because if $\vec{A} = \nabla\phi$, then

$$\text{curl}\vec{A} = \vec{\nabla} \times \vec{\nabla}\phi = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial \phi}{\partial x} & \frac{\partial \phi}{\partial y} & \frac{\partial \phi}{\partial z} \end{vmatrix}$$

$$= \hat{i}\left(\frac{\partial^2\phi}{\partial y\partial z} - \frac{\partial^2\phi}{\partial z\partial y}\right) + \hat{j}\left(\frac{\partial^2\phi}{\partial z\partial x} - \frac{\partial^2\phi}{\partial x\partial z}\right) + \hat{k}\left(\frac{\partial^2\phi}{\partial x\partial y} - \frac{\partial^2\phi}{\partial y\partial x}\right) = 0 \qquad \ldots(7)$$

Thus, if $\vec{A}$ is gradient of a scalar function, the curl of $\vec{A}$ vanishes. Such a vector, whose curl or rotation is zero, is called irrotational vector:

The line integral of any conservative force $\vec{F}$ along some closed curve of the field is zero and it can be represented by negative gradient of potential energy U, i.e., $\vec{F} = -\vec{\nabla}U$. The value of curl of a conservative force is zero, i.e., curl $\vec{F}$ has value zero and on account of this, conservative force will be irrotational. In other words the condition for conservative force (or field) is that curl $\vec{F} = \vec{0}$.

SOLVED EXAMPLES

Example 1: *Find grad ϕ at the point $(-1,-2,1)$, where $\phi = x^2y + .x$.*

Solution: Given, grad $\phi = \nabla\phi = \hat{i}\frac{\partial\phi}{\partial x} + \hat{j}\frac{\partial\phi}{\partial y} + \hat{k}\frac{\partial\phi}{\partial z}$

Now, $\phi = x^2y + xz$

$\therefore$ $\frac{\partial\phi}{\partial x} = 2xy + z; \frac{\partial\phi}{\partial y} = x^2; \frac{\partial\phi}{\partial z} = x$

$\therefore$ $\nabla\phi = (2xy + z)\hat{i} + x^2\hat{j} + x\hat{k}$

Hence, $\nabla\phi$ at the point $(-1,-2,1) = 5\hat{i} + \hat{j} - \hat{k}$

[By substituting $x = -1$; $y = -2$ and $z = 1$ in eqn. (i)]

Example 2: *Find grad* log$r$, *where r is a position vector.*

Solution: grad log$r$ $= \nabla \log r$

$$= \nabla\log\left(x^2 + y^2 + z^2\right)^{\frac{1}{2}} \qquad \left[\because r = \left(x^2 + y^2 + z^2\right)^{\frac{1}{2}}\right]$$

$$= \frac{1}{2}\nabla\log\left(x^2 + y^2 + z^2\right)$$

$$= \frac{1}{2}\left[\hat{i}\frac{\partial}{\partial x}\log\left(x^2 + y^2 + z^2\right) + \hat{j}\frac{\partial}{\partial y}\log\left(x^2 + y^2 + z^2\right)\right.$$

$$\left. + \hat{k}\frac{\partial}{\partial z}\log\left(x^2 + y^2 + z^2\right)\right]$$

$$= \frac{1}{2}\hat{i}\frac{2x}{\left(x^2 + y^2 + z^2\right)} + \frac{1}{2}\hat{j}\frac{2y}{x^2 + y^2 + z^2} + \frac{1}{2}\hat{k}\frac{2z}{x^2 + y^2 + z^2}$$

$$= \frac{1}{x^2+y^2+z^2}\left(\hat{i}x+\hat{j}y+\hat{k}z\right) = \frac{r}{r^2}.$$

Example 3: *Evaluate grad* $\left(\frac{1}{r}\right)$, *where* r *is a position vector.*

Solution: $\text{grad}\left(\frac{1}{r}\right) = \nabla\frac{1}{r} = \nabla\frac{1}{\sqrt{x^2+y^2+z^2}} = \nabla\left(x^2+y^2+z^2\right)^{-\frac{1}{2}}$

$$= \hat{i}\frac{\partial}{\partial x}\left(x^2+y^2+z^2\right)^{-\frac{1}{2}} + \hat{j}\frac{\partial}{\partial y}\left(x^2+y^2+z^2\right)^{-\frac{1}{2}}3$$

$$+\hat{k}\frac{\partial}{\partial z}\left(x^2+y^2+z^2\right)^{-\frac{1}{2}}$$

$$= \hat{i}\left[-\frac{1}{2}\left(x^2+y^2+z^2\right)^{\frac{-3}{2}}\cdot 2x\right] + \hat{j}\left[-\frac{1}{2}\left(x^2+y^2+z^2\right)^{-\frac{3}{2}}\cdot 2y\right]$$

$$+\hat{k}\left[-\frac{1}{2}\left(x^2+y^2+z^2\right)^{-\frac{3}{2}}\cdot 2z\right]$$

$$= -\frac{1}{\left(x^2+y^2+z^2\right)^{\frac{3}{2}}}\left(\hat{i}x+\hat{j}y+\hat{k}z\right) = \frac{-\vec{r}}{r^3}.$$

Example 4: Evaluate grad r^n, where $\vec{r}$ is a position vector.

Solution: $\text{grad}\, r^n = \nabla r^n$

$$= \hat{i}\frac{\partial}{\partial x}\left(x^2+y^2+z^2\right)^{\frac{n}{2}} + \hat{j}\frac{\partial}{\partial y}\left(x^2+y^2+z^2\right)^{\frac{n}{2}} + \hat{k}\frac{\partial}{\partial z}\left(x^2+y^2+z^2\right)^{\frac{n}{2}}$$

$$= \hat{i}\left[\frac{n}{2}\left(x^2+y^2+z^2\right)^{\frac{n}{2}-1}2x\right] + \hat{j}\left[\frac{n}{2}\left(x^2+y^2+z^2\right)^{\frac{n}{2}-1}2y\right]$$

$$+\hat{k}\left[\frac{n}{2}\left(x^2+y^2+z^2\right)^{\frac{n}{2}-1}2z\right]$$

$$= n\left(x^2+y^2+z^2\right)^{\frac{n}{2}-1}\left[\hat{i}x+\hat{j}y+\hat{k}z\right] = n\left(r^2\right)^{\frac{n}{2}-1}\vec{r} = nr^{n-2}\vec{r}$$

Example 5: *Evaluate grad* r, *where* $r = \left(x^2+y^2+z^2\right)^{\frac{1}{2}}$.

Solution: From example 2.4, grad $r^n = nr^{n-2}\vec{r}$

Putting $n = 1$, we have

$$\text{grad}\, r = 1(r)^{1-2}\vec{r} = r^{-1}\vec{r} = \frac{\vec{r}}{r} = \hat{r}$$

Alternative Method:

$$\text{grad}\, r = \nabla r = \hat{i}\frac{\partial r}{\partial x} + \hat{j}\frac{\partial r}{\partial y} + \hat{k}\frac{\partial r}{\partial z}$$

$$= \hat{i}\frac{\partial}{\partial x}\left(x^2+y^2+z^2\right)^{\frac{1}{2}} + \hat{j}\frac{\partial}{\partial y}\left(x^2+y^2+z^2\right)^{\frac{1}{2}} + \hat{k}\frac{\partial}{\partial z}\left(x^2+y^2+z^2\right)^{\frac{1}{2}}$$

$$= \hat{i}\frac{1}{2}\left(x^2+y^2+z^2\right)^{-\frac{1}{2}}2x + \hat{j}\frac{1}{2}\left(x^2+y^2+z^2\right)^{-\frac{1}{2}}2y + \hat{k}\frac{1}{2}\left(x^2+y^2+z^2\right)^{-\frac{1}{2}}2z$$

$$= \left(x^2+y^2+z^2\right)^{-\frac{1}{2}}\left[\hat{i}x+\hat{j}y+\hat{k}z\right] = \frac{1}{r}\vec{r} = \frac{1}{r}(r\hat{r}) = \hat{r}$$

Example 6: *Find a unit vector normal to the surface* $z = x^2 + y^2$ *at the point* $(1,2,5)$.

Solution: We know that vector grad u is perpendicular to level surface.

Given surface is $z = x^2 + y^2$ or $x^2 + y^2 - z = 0$

Example 7: *If volume* V *is enclosed by a surface* S, *then find the value of* $\iint_S \vec{F}\cdot d\vec{s}$ *for the vector field* $\vec{F} = 3x\hat{i} + 4y\hat{j} + 2z\hat{k}$

Solution: By Gauss' theorem,

$$\iint_S \vec{F}\cdot d\vec{s} = \iiint_V \left(\text{dir v}\,\vec{F}\right)dV$$

Here,
$$\text{div}\vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

$$= \frac{\partial}{\partial x}(3x) + \frac{\partial}{\partial y}(4y) + \frac{\partial}{\partial z}(2z)$$

$$= 3+4+2 = 9$$

Hence,
$$\iint_S \vec{F}\cdot d\vec{s} = \iiint_V 9dV = 9V$$

Example 8: Prove that $\iint_S \vec{r}\cdot d\vec{s} = 3V$ whose, $\vec{r}$ is the position vector at any point on surface enclosing volume V.

Solution: By Gauss' theorem,

$$\iint_S \vec{A}\cdot d\vec{s} = \iiint_V \left(\text{div}\vec{A}\right)dV$$

Here, $\vec{A} = \vec{r}$ and $\text{div}\vec{A} = \text{div}\,,\ \vec{r} = 3$

So,
$$\iint_S \vec{r}\cdot d\vec{s} = \iiint_V \text{div}\,\vec{r}\,dV$$

$$= \iiint_V 3dV$$

$$= 3V$$

Example 9: A rigid body is making revolutions with velocity 2 radian per second about the axis $\left(3\hat{j}-\hat{k}\right)$. This axis passes through the point $\left(\hat{i}+3\hat{j}-\hat{k}\right)$. Find the velocity of the particle at point $\left(4\hat{i}-2\hat{j}+\hat{k}\right)$.

Solution: If the position vector of the particle with respect to the given point is $\vec{r}$, then

$$\vec{r} = \left(4\hat{i}-2\hat{j}+\hat{k}\right)-\left(\hat{i}+3\hat{j}-\hat{k}\right)$$

$$= 3\hat{i}-5\hat{j}+2\hat{k}$$

Angular velocity with respect to axis $\left(3\hat{j}-\hat{k}\right)$

$\vec{\omega}$ = (Magnitude of angular Velocity)

(Unit vector in its direction)

$$= 2\frac{3\hat{j}-\hat{k}}{\left|3\hat{j}-\hat{k}\right|} = \frac{2}{\sqrt{10}}\left(3\hat{j}-\hat{k}\right)$$

$\therefore$ $\vec{v} = \vec{\omega}\times\vec{r}$

$$= \frac{2}{\sqrt{10}}\left(3\hat{j}-\hat{k}\right)\times\left(3\hat{i}-5\hat{j}+2\hat{k}\right)$$

$$= \frac{2}{\sqrt{10}}\left(\hat{i}-3\hat{j}-9\hat{k}\right)$$

$$\text{Its Magnitude} = \frac{2}{\sqrt{10}}\sqrt{91}$$

$$= 6 \text{ units (approx)}$$

Example 10: *If* $\vec{A} = -y\hat{i}+x\hat{j}$, *then find (i)* curl$\vec{A}$ *and (ii) line integral* $\oint \vec{A}\cdot\vec{dl}$ *for the closed curve* $x^2+y^2=r^2, z=0$. *Hence verify Stroke's theorem.*

Solution: (i) $A_1 = -y, A_y = x, A_z = 0$

$\therefore$ $\frac{\partial A_x}{\partial y} = -1, \frac{\partial A_y}{\partial x} = 1$

and the other derivatives are zero.

So, $$\text{curl}\vec{A} = \left(\frac{\partial A_y}{\partial x}-\frac{\partial A_x}{\partial y}\right)\hat{k}$$

$$= (1+1)\hat{k} = 2\hat{k}$$

$$\oint \vec{A}\cdot\vec{dl} = \int\left(-y\hat{i}+x\hat{j}\right)\cdot\left(dx\hat{i}+dy\hat{j}\right)$$

$$= \int -ydx + \int xdy$$

For circle $x^2 + y^2 = r^2$

and $x = r\cos\theta,\ y = r\sin\theta$

$\therefore$ $dx = -r\sin\theta d\theta, dy = r\cos\theta d\theta$

$\therefore$ $\oint \vec{A}\cdot\vec{dl} = \int_0^{2\pi} r^2\left(\cos^2\theta + \sin^2\theta\right)d\theta = 2\pi r^2$

Now, according Stroke's theorem,

$$\oint \vec{A}\cdot\vec{dl} = \iint_S \text{curl}\vec{A}\cdot d\vec{s} = \iint_S 2k\cdot ds\hat{k}$$

$$= \iint_S 2ds = 2S = 2\pi r^2$$

Hence, Stroke's theorem is verified.

Example 11: *If* $\vec{r}$ *is the position vector of any point, prove that*

$$\text{grad}\, r^n \text{ or } \vec{\nabla} r^n = nr^{n-2}\vec{r}$$

where $\vec{r} = \hat{i}x + \hat{j}y + \hat{k}z$.

Solution: $r^2 = x^2 + y^2 + z^2$ or $r = \left(x^2 + y^2 + z^2\right)^{1/2}$

Now, $\text{grad}\phi = \hat{i}\frac{\partial\phi}{\partial x} + \hat{j}\frac{\partial\phi}{\partial y} + \hat{k}\frac{\partial\phi}{\partial z}$

$\therefore$ $\text{grad}r^n = \hat{i}\frac{\partial}{\partial x}\left(x^2 + y^2 + z^2\right)^{\frac{n}{2}} + \hat{j}\frac{\partial}{\partial y}\left(x^2 + y^2 + z^2\right)^{\frac{n}{2}} + \hat{k}\frac{\partial}{\partial z}\left(x^2 + y^2 + z^2\right)^{\frac{n}{2}}$

$$= \hat{i}\frac{n}{2}\left(x^2 + y^2 + z^2\right)^{\frac{n}{2-1}}\cdot 2x + \ldots$$

$$= nr^{n-2}\left(\hat{i}x + \hat{j}y + \hat{k}z\right) = nr^{n-2}\vec{r}$$

Example 12: *Prove that div* $\left(\vec{A}\times\vec{B}\right) = \vec{B}\cdot\text{curl}\vec{A} - \vec{A}\cdot\text{curl}\vec{B}$.

Solution: Let $\left(\vec{A}\times\vec{B}\right)$ be equal to $\vec{V}$. Then

$$\vec{V} = \vec{A}\times\vec{B} = \left(A_yB_z - A_zB_y\right)\hat{i} + \left(A_zB_x - A_xB_z\right)\hat{j} + \left(A_xB_y - A_yB_x\right)\hat{k}$$

$$= V_x\hat{i} + V_y\hat{j} + V_z\hat{k}$$

Now, $\text{div}\left(\vec{A}\times\vec{B}\right) = \text{div}\vec{V} = \nabla\cdot\vec{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$

$$= \frac{\partial}{\partial x}\left(A_yB_z - A_zB_y\right) + \frac{\partial}{\partial y}\left(A_zB_x - A_xB_z\right) + \frac{\partial}{\partial z}\left(A_xB_y - A_yB_x\right)$$

$$= \left[B_x\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + B_y\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + B_z\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\right]$$

$$-\left[A_x\left(\frac{\partial B_z}{\partial y}-\frac{\partial B_y}{\partial z}\right)-A_y\left(\frac{\partial B_x}{\partial z}-\frac{\partial B_z}{\partial x}\right)-A_z\left(\frac{\partial B_y}{\partial x}-\frac{\partial B_x}{\partial y}\right)\right]$$

$$=\vec{B}\cdot\text{curl}\vec{A}-\vec{A}\cdot\text{curlB}.$$

Example 13: *If* $\vec{r}$ *is the position vector of a point, deduce the value of grad* $\left(\frac{1}{r}\right)$.

Solution: $\text{grad}\left(\frac{1}{r}\right)=\text{grad}\left(r^{-1}\right)$

Substituting $r=\sqrt{x^2+y^2+z^2}$ and solving as above. we can get

$$\text{grad}\left(\frac{1}{r}\right)=(-1)r^{-1-2}\vec{r}=-\frac{r}{r^3}=-\frac{\vec{r}}{r^2}$$

Example 14: *Prove that* $\left[\text{grad}\,f(r)\right]\times\vec{r}=0$

Solution: $\text{grad}f(r)=\frac{\partial f}{\partial r}\hat{r}=\frac{\partial f}{\partial r}\cdot\frac{\vec{r}}{r}$

$\therefore\quad\left[\text{grad}f(r)\right]\times\vec{r}=\frac{\partial f}{\partial r}\cdot\frac{\vec{r}}{r}\times\vec{r}=0$ [$\because$ cross multiplication of equal vectors is zero]

Example 15: *Find the set of vectors reciprocal to* $\left(2\hat{i},3\hat{j},4\hat{k}\right)$.

Solution: $a^*=\frac{b\times c}{a\cdot b\times c}=\frac{3\hat{j}\times4\hat{k}}{2\hat{i}\cdot\left(3\hat{j}\times4\hat{k}\right)}=\frac{12\hat{i}}{24}=\frac{\hat{i}}{2}$

$$b^*=\frac{c\times a}{a\cdot b\times c}=\frac{4\hat{k}\times2\hat{i}}{24}=\frac{\hat{j}}{3}$$

$$c^*=\frac{a\times b}{a(b\times c)}=\frac{2\hat{i}\times3\hat{j}}{24}=\frac{\hat{k}}{4}$$

Thus, the set of reciprocal vectors is $\left(\frac{\hat{i}}{2},\frac{\hat{j}}{3},\frac{\hat{k}}{4}\right)$.

Example 16: *Prove that* $\text{curlcurl}\vec{A}=\text{grad}\,\text{div}\vec{A}-\vec{\nabla}^2\vec{A}$

Solution: $\text{curlcurl}\vec{A}=\vec{\nabla}\times\left(\vec{\nabla}\times\vec{A}\right)$

We know that $\vec{A}\times\left(\vec{B}\times\vec{C}\right)=\vec{B}\left(\vec{A}\cdot\vec{C}\right)-\left(\vec{A}\cdot\vec{B}\right)\vec{C}$

$\therefore\qquad\text{curlcurl}\vec{A}=\vec{\nabla}\left(\vec{\nabla}\cdot\vec{A}\right)-\left(\vec{\nabla}\cdot\vec{\nabla}\right)\vec{A}$

$$=\text{grad}\,\text{div}\vec{A}-\overrightarrow{\nabla^2}\vec{A}$$

Example 17: *If* ϕ *and* ψ *are differential function of* x,y,z, *then prove that grad*

$$(\phi\psi)=\phi\vec{\nabla}\psi+\psi/\vec{\nabla}\phi.$$

Solution: $\text{grad}(\phi\psi) = \left(\hat{i}\frac{\partial}{\partial x}+\hat{j}\frac{\partial}{\partial y}+\hat{k}\frac{\partial}{\partial z}\right)(\phi\psi)$

$$= \hat{i}\left(\phi\frac{\partial\psi}{\partial x}+\psi\frac{\partial\phi}{\partial x}\right)+\hat{j}\left(\phi\frac{\partial\psi}{\partial y}+\psi\frac{\partial\phi}{\partial y}\right)+\hat{k}\left(\phi\frac{\partial\psi}{\partial z}+\psi\frac{\partial\phi}{\partial z}\right)$$

$$= \phi\left(\hat{i}\frac{\partial\psi}{\partial x}+\hat{j}\frac{\partial\psi}{\partial y}+\hat{k}\frac{\partial\psi}{\partial z}\right)+\psi\left(\hat{i}\frac{\partial\phi}{\partial x}+\hat{j}\frac{\partial\phi}{\partial y}+\hat{k}\frac{\partial\phi}{\partial z}\right)$$

$$= \phi\vec{\nabla}\psi+\psi+\vec{\nabla}\phi$$

Example 18: *If* $\vec{r} = x\hat{i}+y\hat{j}+z\hat{k}$ *then find the value of div* $\vec{r}$ *.*

Solution: $div\vec{r} = \vec{\nabla}\cdot\vec{r}$

$$= \left(\hat{i}\frac{\partial}{\partial x}+\hat{j}\frac{\partial}{\partial y}+\hat{k}\frac{\partial}{\partial z}\right)\cdot\left(x\hat{i}+y\hat{j}+z\hat{k}\right)$$

$$= \frac{\partial x}{\partial x}+\frac{\partial y}{\partial y}+\frac{\partial z}{\partial z} = 1+1+1 = 3$$

Example 19: *If* $\vec{A} = (x+2y)\hat{i}+(y-z)\hat{j}+(x-cz)\hat{k}$ *is a solenoidal vector, then find the value of* c *.*

Solution: Here, $A_x = x+2y$

$$A_y = y-z$$

$$A_z = x-cz$$

$\therefore$ $\frac{\partial A_x}{\partial x} = 1, \frac{\partial A_y}{\partial y} = 1, \frac{\partial A_x}{\partial x}+\frac{\partial A_y}{\partial y}+\frac{\partial A_z}{\partial z} = -c$

So, $\vec{\nabla}\cdot\vec{A} = \frac{\partial A_x}{\partial x}+\frac{\partial A_y}{\partial y}+\frac{\partial A_z}{\partial z} = 1+1-c = 2-c$

But $\vec{A}$ is a solenoidal vector

$\therefore$ $\vec{V}\cdot\vec{A} = 0$

$\therefore$ $2-c = 0$

$\Rightarrow$ $c = 2$

Example 20: *Prove that the vector* $\vec{A} = \hat{i}\,yz+\hat{j}\,xz+\hat{k}\,xr$ *is solenoidal and proportional*

Solution: Here, $\vec{A} = \hat{i}yz+\hat{j}xz+\hat{k}xy$

$$A_x = yz$$

and $A_y = xz$

$$\frac{\partial A_x}{\partial x} = 0, \frac{\partial A_y}{\partial y} = 0 \text{ and } \frac{\partial A_z}{\partial z} = 0$$

$$\therefore \quad \vec{\nabla} \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

Hence $\vec{A}$ is a solenoidal vector.

Again, $$\text{curl}\,\hat{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_Y & A_Z \end{vmatrix} - \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz & xz & xy \end{vmatrix}$$

$$= \hat{i}(x - x) + \hat{j}(y - y) + \hat{k}(z - z)$$

$$= 0$$

Hence $\vec{A}$ is irritational.

Example 21: *If $\vec{A}$ and $\vec{B}$ an irrotational vectors, then prove that $\vec{A} \times \vec{B}$ is a solenoidal*

Solution: Given: $\vec{V} \times \vec{A} = 0$ and $\vec{V} \times \vec{B} = 0$

We know that

$$\vec{V} \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot \text{curl}\,\vec{A} - \vec{A} \cdot \text{curl}\,\vec{B}$$

$$= \vec{B} \cdot (\vec{V} \times \vec{A})\ \vec{A} \cdot (\vec{V} \times \vec{B})$$

$$= \vec{B} \cdot \vec{0} - \vec{A} \cdot \vec{0}$$

$$= 0 - 0 = 0$$

Hence $\vec{A} \times \vec{B}$ is a solenoidal vector.

Example 22: *Prove that $\vec{F} = a(x\hat{i} + y\hat{j})$ is a conservative vector field.*

Solution: For a conservative field the value of $\text{curl}\,\vec{F}$ be zero. Here.

$$\text{curl}\,\vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ ax & az & 0 \end{vmatrix}$$

$$= \hat{i}\left[\frac{\partial}{\partial y}(0) - \frac{\partial}{\partial z}(ay)\right] + \hat{j}\left[\frac{\partial}{\partial z}(ax) - \frac{\partial}{\partial z}(0)\right]$$

$$+ \hat{k}\left[\frac{\partial}{\partial x}(ay) - \frac{\partial}{\partial y}(ax)\right]$$

$$= \hat{i}(0 - 0) + \hat{j}(0 - 0) + \hat{k}(0 - 0)$$

$$= 0$$

Hence $\vec{F}$ is a conservative field.

Example 23: *If $\vec{a}$ is a constant vector and $\vec{r}$ is positive vector, then prove that*

(i) $\text{div}(\vec{a}\times\vec{r})=0$

(ii) $\text{curl}(\vec{a}\times\vec{r})=2\vec{a}$

Solution: (i) In component form, let the constant vector $\vec{a}=a_x\hat{i}+a_y\hat{j}+a_z\hat{k}$ and $\vec{r}=x\hat{i}+y\hat{j}+z\hat{k}$.

$$\therefore \quad \text{div}(\vec{a}\times\vec{r}) = \vec{\nabla}\cdot(\vec{a}\times\vec{r})$$

$$= \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_x & a_y & a_z \\ x & y & z \end{vmatrix}$$

$$= \frac{\partial}{\partial x}(a_yz-a_zy)+\frac{\partial}{\partial y}(a_zx-a_z)+\frac{\partial}{\partial z}(a_xy-a_yx)$$

$$= 0$$

$[\because a_x, a_y, a_z$ are constants and x, y, z are independent of each other.]

(ii) Given $\vec{a}\times\vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_x & a_y & a_z \\ x & y & z \end{vmatrix}$

$$= (a_yz-a_zy)\hat{i}+(a_zx-a_xz)\hat{j}+(a_xy-a_yx)\hat{k}$$

$$\therefore \quad \text{curl}(\vec{a}\times\vec{r}) = \vec{\nabla}\times\vec{a}\times\vec{r}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_yz-a_zy & a_zx-a_xz & a_xy-a_yx \end{vmatrix}$$

$$= \hat{i}\left[\frac{\partial}{\partial y}(a_xy-a_yx)-\frac{\partial}{\partial z}(a_zx-a_xz)\right]+\ldots$$

$$= \hat{i}(a_x+a_x)+\hat{j}(a_y+a_yz)+\hat{k}(a_z+a_z)$$

$$= 2(a_x\hat{i}+a_y\hat{j}+a_z\hat{k})$$

$$= 2\cdot\vec{a}$$

Example 24: *Prove that curl grad $\phi=0$.*

Solution: Curl grad $\phi = \vec{\nabla} \times \vec{\nabla}\phi$

$$= \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right) \times \left(\hat{i}\frac{\partial \phi}{\partial x} + \hat{j}\frac{\partial \phi}{\partial y} + \hat{k}\frac{\partial \phi}{\partial z}\right)$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial \phi}{\partial x} & \frac{\partial \phi}{\partial y} & \frac{\partial \phi}{\partial z} \end{vmatrix}$$

$$= \hat{i}\left(\frac{\partial^2 \phi}{\partial y \partial z} - \frac{\partial^2 \phi}{\partial z \partial y}\right) + \hat{j}\left(\frac{\partial^2 \phi}{\partial z \partial x} - \frac{\partial^2 \phi}{\partial x \partial z}\right) + \hat{k}\left(\frac{\partial^2 \phi}{\partial x \partial y} - \frac{\partial^2 \phi}{\partial y \partial x}\right)$$

$$= 0 \qquad \left[\because \frac{\partial^2 \phi}{\partial y \partial z} = \frac{\partial^2 \phi}{\partial z \partial y}; \frac{\partial^2 \phi}{\partial z \partial x} = \frac{\partial^2 \phi}{\partial x \partial z}; \frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x}\right]$$

Example 25: *Prove that div curl* $\vec{A} = 0$.

Solution: divcurl$\vec{A} = \vec{\nabla} \cdot \left(\vec{\nabla} \times \vec{A}\right)$

$$= \vec{\nabla} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

$$= \vec{\nabla}\left[\hat{i}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \hat{j}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + \hat{k}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\right]$$

$$= \frac{\partial}{\partial x}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \frac{\partial}{\partial y}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + \frac{\partial}{\partial z}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)$$

$$= \frac{\partial^2 A_z}{\partial x \partial y} - \frac{\partial^2 A_y}{\partial x \partial z} + \frac{\partial^2 A_x}{\partial y \partial z} - \frac{\partial^2 A_z}{\partial y \partial x} + \frac{\partial^2 A_y}{\partial z \partial x} - \frac{\partial^2 A_x}{\partial z \partial y} = 0$$

Example 26: *If* $\vec{a}$ *is a constant vector and* $\vec{r}$ *is the positive vector, then prove that* $\text{grad}\left(\vec{a} \cdot \vec{r}\right) = \vec{a}$

Solution: Let, $\vec{a} = a_x\hat{i} + a_y\hat{j} + a_z\hat{k}$

and $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

Then, $\vec{a} \cdot \vec{r} = a_x x + a_y y + a_z z$

$$\therefore \quad \text{grad}(\vec{a}\cdot\vec{r}) = \hat{i}\frac{\partial}{\partial x}(a_x x + a_y y + a_z z) + \hat{j}\frac{\partial}{\partial y}(a_x x + a_y y + a_z z)$$

$$+\hat{k}\frac{\partial}{\partial z}(a_x x + a_y y + a_z z)$$

$$= \hat{i}(a_x \cdot 1 + 0 + 0) + \hat{j}(0 + a_y \cdot 1 + 0) + \hat{k}(0 + 0 + a_z \cdot 1)$$

$$= a_1\hat{i} + a_1\hat{j} + a_1\hat{k}$$

QUESTIONS

MULTIPLE CHOICE QUESTIONS

1. If ϕ is scalar field then gradϕ is :

(a) Scalar
(b) Vector
(c) Zero
(d) Either vector or scalar

2. If ϕ is gravitational potential at a point, then $\vec{F}$ will be the intensity of the gravitation field at the place (point) where:

(a) $\vec{F} = \text{curl}\phi$
(b) $\vec{F} = \vec{\nabla}\phi$
(c) $\vec{F} = -\nabla\phi$
(d) $\vec{F} = \text{div}\phi$

3. The divergence of a vector is:

(a) Positive or negative
(b) Positive or zero
(c) Always positive
(d) Positive or negative or zero

4. The divergence of r is :

(a) $x + y + z$
(b) r
(c) 3
(d) i + j + k

5. The line integral of a vector between two points in a conservative vector lich depends upon:

(a) The displacement between the points
(b) Positions of the points
(c) Both (a) and (b) are correct
(d) Both (a) and (b) are incorrect

6. The value of $\vec{\nabla}\left(\frac{1}{r}\right)$ is:

(a) $-\frac{\vec{r}}{r^3}$
(b) $\vec{r}$
(c) 0
(d) r^2

7. Grad $(\vec{r}\cdot\vec{r})$ is:

(a) $\vec{r}$
(b) $2\vec{r}$
(c) $\frac{\vec{r}}{2}$
(d) $2(x + y + z)$

8. If $\vec{A} = \text{grad}\phi$, then $\vec{A}$ is:

(a) Rotational vector
(b) Irrotational vector
(c) Solenoidal vector
(d) Non-solenoidal

9. Curl of vector function at a point is:

(a) A vector
(b) A scalar
(c) Depends on the position vector of the point
(d) An exponential function

10. If $\oint \vec{A} \cdot d\vec{l} = 0$ for a closed curve, then:

(a) $\vec{\nabla} \times \vec{A} = \vec{0}$
(b) $\vec{\nabla} \cdot \vec{A} = 0$
(c) $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0$
(d) $\vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{0}$

11. If $\vec{F} = (x^2 - z^2)\hat{i} + 2\hat{j} + 2xz\hat{k}$, then divF is :

(a) $2x$
(b) 0
(c) $4x$
(d) $2x + 2z$

12. If $\vec{A}$ is a vector field, then:

(a) $\text{curl}\vec{A} = \vec{\nabla} \cdot \vec{A}$
(b) $\text{curl}\vec{A} = \vec{\nabla}\vec{A}$
(c) $\text{curl}\vec{A} = \vec{\nabla} \times \vec{A}$
(d) $\text{curl}\vec{A} = \vec{0}$

13. If A is vertical upward and $\vec{B}$ is along north, then the direction of $\vec{A} \times \vec{B}$ is:

(a) West
(b) East
(c) Vertically downwards
(d) South

14. Surface area $\vec{S} = 5\hat{k}$ lies in:

(a) $X - Y$ plane
(b) I -Z plane
(c) $Z - X$ plane
(d) along Z -axis

15. The value of $\int_{-\infty}^{\infty} f(t)\delta(t)dt$ is:

(a) $f(1)$
(b) $f(\infty)$
(c) $f(0)$
(d) $f(t)$

16. If $\delta(x)$ is an even function of x, then:

(a) $\delta(-x) = \delta(x)$
(b) $\delta(-x) = 0$
(c) $\delta(x) = -x^2$
(d) $\delta(x) = 0$

17. The value of $x\delta(x)$ is:

(a) 1
(b) ∞
(c) 0
(d) x

18. The value of $\int_{-\infty}^{\infty} \delta(x)dx$ is:

(a) 0
(b) 1
(c) -1
(d) ∞

19. The value of $\delta(-x)$ is:

(a) 0 (b) $-\delta(x)$

(c) $\delta(x)$ (d) ∞

20. The value of $\int_{-\infty}^{\infty} f(x)\delta(x-a)dx$ is

(a) $f(x)$ (b) $f(a)$

(c) $f(0)$ (d) $f(\infty)$

21. The expression $x\dfrac{d\delta(l)}{dx}$ is equal to

(a) $\delta(x)$ (b) $\delta(-x)$

(c) $-\delta(x)$ (d) 0

LONG ANSWER QUESTIONS

1. Explain curl, divergence and gradient of a vector. Give the physical interpretation of divergence. State and prove Gauss' theorem
2. State and prove Stroke's theorem.
3. What is the gradient of a scalar function ? Show that the vector $\nabla\phi$ is normal to the surface $\phi =$ constant.
4. Give the physical interpretation of
   (a) Define flux of a vector field.
   (b) What do you mean by curl of a vector field? Prove that curl $V = \vec{\nabla}\times\vec{V}$ where the symbols have their usual meaning.
5. If a vector field may be expressed as $\vec{A} = \theta\nabla\phi$, where θ and ϕ being the scalar functions of position, show with the help of Gauss' theorem that
$$\iiint_V \left(\theta\nabla^2\phi + \nabla\theta\cdot\nabla\phi\right)dV = \iint_S (\theta\nabla\phi)\cdot d\vec{s}$$
6. What is Dirac Delta function? Show that
$$\delta(x) = \delta(-x)$$

SHORT ANSWERS QUESTIONS

1. What do you understand by gradient of a scalar function?
2. Is the gradient of scalar function a vector?
3. What is the meaning of potential and temperature gradients?
4. Give physical interpretation of :
   (a) Gradient of a scalar field.
   (b) Divergence and curl of a vector field.
5. If $\vec{F} = 3x\hat{i} + 4y\hat{j} + 2z\hat{k}$, find div$\vec{F}$. [Ans. 9.]

6. Divergence of a vector is scalar and the gradient of it scalar is vector. Explain them with examples.
7. What is the meaning of $\vec{V}\cdot\vec{A}=0$, where $\vec{A}$ is a vector.

   [Ans. The meaning of $\vec{V}\cdot\vec{A}=0$, is that flux lines of vector $\vec{A}$ are closed.]
8. Define curt of a vector. Give its physical significance.
9. What is the divergence of a vector scalar or vector?
10. What is a solenoidal vector? Give example.
11. State Gauss' and Stroke's theorems.
12. Prove that uniform electric field is irrotational.

    [**Ans:** The uniform electric field is expressed as

    $\vec{E} = -\vec{\nabla}V$ therefore curl $\vec{E} = \vec{\nabla}\times\left(-\vec{\nabla}V\right)=\vec{0}$

    Therefor $\vec{E}$ is an irrational filed]
13. Prove that curl of a conservative field is zero.

    [***Hint:*** Conservative field is expressed in terms of the gradient of a scalar field and curl of a gradient is zero]
14. Give an example of irrotational field.
15. Prove that curl grad $\phi=0$ or $\vec{\nabla}\times\vec{\nabla}\phi=\vec{0}$.
16. State Helmholtz's theorem.

MULTIPLE CHOICE QUESTIONS ANSWERS

1	(b)	**2**	(c)	**3**	(d)	**4**	(c)	**5**	(b)
6	(a)	**7**	(b)	**8**	(b)	**9**	(a)	**10**	(a)
11	(c)	**12**	(c)	**13**	(a)	**14**	(a)	**15**	(c)
16	(a)	**17**	(c)	**18**	(b)	**19**	(c)	**20**	(b)
21	(c)								

Example 8: *Show that*

(i) $\left(\vec{A}\times\vec{B}\right)\times\left(\vec{C}\times\vec{D}\right)=\left[\vec{A}\cdot\left(\vec{B}\times\vec{D}\right)\right]\vec{C}-\left[\vec{A}\cdot\left(\vec{B}\times\vec{C}\right)\right]\vec{D}$

(ii) $\vec{A}\times\left[\vec{B}\times\left(\vec{C}\times\vec{D}\right)\right]=\left(\vec{A}\times\vec{C}\right)\left(\vec{B}\cdot\vec{D}\right)-\left(\vec{A}\times\vec{D}\right)\left(\vec{B}\cdot\vec{C}\right).$

Solution: **(i)** If we put $\vec{A}\times\vec{B}=\vec{V}$, then

$$\vec{V}\times\left(\vec{C}\times\vec{D}\right)=\left(\vec{V}\cdot\vec{D}\right)\vec{C}-\left(\vec{V}\cdot\vec{C}\right)\vec{D}$$

$$=\left[\left(\vec{A}\times\vec{B}\right)\cdot\vec{D}\right]\vec{C}-\left[\left(\vec{A}\times\vec{B}\right)\cdot\vec{C}\right]\vec{D}$$

As in a scalar triple product like $\left[\left(\vec{A}\times\vec{B}\right)\cdot\vec{D}\right]$, the positions of dot and cross may be interchanged.

$$\therefore\quad\left(\vec{A}\times\vec{B}\right)\times\left(\vec{C}\times\vec{D}\right)=\left[\vec{A}\cdot\left(\vec{B}\times\vec{D}\right)\right]\vec{C}-\left[\vec{A}\cdot\left(\vec{B}\times\vec{C}\right)\right]\vec{D}.$$

(ii) $\vec{A}\times\left[\vec{B}\times\left(\vec{C}\times\vec{D}\right)\right]=\vec{A}\times\left[\left(\vec{B}\cdot\vec{D}\right)\vec{C}-\left(\vec{B}\cdot\vec{C}\right)\vec{D}\right]$

$$=\left(\vec{A}\times\vec{C}\right)\left(\vec{B}\cdot\vec{D}\right)-\left(\vec{A}\times\vec{D}\right)\left(\vec{B}\cdot\vec{C}\right).$$

Example 9: *Evaluate* $\left(\vec{A}\times\vec{B}\right)\cdot\left(\vec{C}\times\vec{D}\right)$.

Solution: Let $\vec{A}\times\vec{B}=\vec{x}$

Then $\left(\vec{A}\times\vec{B}\right)\cdot\left(\vec{C}\times\vec{D}\right)=\vec{x}\cdot\left(\vec{C}\times\vec{D}\right)=\left(\vec{x}\times\vec{C}\right)\cdot\vec{D}$

$$=\left[\left(\vec{A}\times\vec{B}\right)\times\vec{C}\right]\cdot\vec{D}$$

$$=\left[\left(\vec{A}\cdot\vec{C}\right)\vec{B}-\left(\vec{A}\cdot\vec{B}\right)\vec{C}\right]\cdot\vec{D}$$

$$=\left(\vec{A}\cdot\vec{C}\right)\left(\vec{B}\cdot\vec{D}\right)-\left(\vec{A}\cdot\vec{B}\right)\left(\vec{C}\cdot\vec{D}\right)$$

1.16 FIELDS

If a physical quantity has a given set of elements in space, and for each element, the quantity varies depending upon the value of the elements; the physical quantity is called function of the elements. If a physical quantity varies from point to point in space, it can be expressed as a continuous function of the position of a point in a region of space, then such a function is called the function of position or point function and the region it specifies the physical quantity is called the field. There are two main kinds of the fields, corresponding to the nature of physical quantities. For instance, at each point in space expressed in terms of position, a temperature, or a pressure, or a density of the air can be specified.

Again, one may have at each point in space a velocity which has both direction and length, or an electric intensity, or a magnetic intensity. Not to multiply examples

unnecessarily, it can be seen that for a given range of points, or lines, or other geometric elements, one may have a set of quantities, corresponding to the various elements of the range, and therefore constituting a function of the range, and these quantities may consist of numerical values, or vectors. When these quantities depends on a simple numerical character they are called scalars, and the function resulting is a **scalar function**. Similarly, if these quantities depend on a vector, the function is called **vector function.** There are mainly two kinds of fields, depending on the nature of the physical quantity

1. Scalar field is defined for scalar functions
2. Vector field is defined for vector functions

1.17 SCALAR FIELD

The scalar field is represented by a continuous scalar function, whose value at each point in space depends on a simple numerical character and is denoted by $\phi(x,y,z)$. The magnitude of such a field does not change abruptly, when it passes from one point to another in its neighborhood. For example, distribution of temperature and density does not vary abruptly in a given region of space. In such cases, one can draw surfaces corresponding to equal values of scalar field, called the level surfaces or lamellar. For the distribution of temperature in a given region of space these level surfaces are called **isothermal** and for density these surfaces are named as **isopycnic** surfaces. Two level surfaces cannot cut each other as if they cut each other, scalar values corresponding to both must hold along their common line, which is contrary to the definition of level surface. Thus scalar point function $\phi(x,y,z)$ is a single valued field and uniform at every point in the region of space.

1.18 VECTOR FIELD

A vector field is represented at every point by a continuous vector function $\vec{F}(x,y,z)$. At any given point of field, the function $\vec{F}(x,y,z)$ is specified by a vector of definite magnitude and direction, both of which change continuously from point to point of field region. Examples of vector fields are: the distribution of velocity in a fluid (liquid or gas), the distribution of electric or magnetic field intensity etc.

Starting from any arbitrary point in the field, proceed an infinitesimal distance in the direction of the vector at that point so as to reach at a closely neighbouring point. Proceeding in a similar way, one obtains in general, a curved line, called the vector line, line of flow or flux line. The tangent at any point of this curved line gives the direction of the vector at that point. For the representation of the magnitude of the vector at any point on a flux line, a small surface perpendicular is drawn and a number of points per unit area are chosen on this surface, which are equal to the magnitude of the vector. Through each of these points flux lines can be drawn. Thus, the field is mapped out by

flux lines. The direction of flux lines is that of vector function, their density, i.e., the number of lines crossing per unit area perpendicular to their direction, is the magnitude of the vector in that direction.

QUESTIONS

MULTIPLE CHOICE QUESTIONS

1. A scalar has:

(a) Magnitude (b) Direction

(c) Both (d) None of these

2. A vector has:

(a) Magnitude (b) No magnitude

(c) Zero magnitude but no specific direction (d) Magnitude as well as direction

3. The displacement is a :

(a) Scalar quantity (b) Vector quantity

(c) Both (d) None of these

4. Three vectors are coplanar if :

(a) They are parallel (b) Non parallel

(c) Lie in same plane (d) None of these

5. Two vectors $\vec{A}$ and $\vec{B}$ are perpendicular if :

(a) $\vec{A}\cdot\vec{B}=0$ (b) $\vec{A}\times\vec{B}=0$

(c) $\vec{A}+\vec{B}=0$ (d) None of these

6. Two vectors $\vec{A}$ and $\vec{B}$ are parallel if :

(a) $\vec{A}\times\vec{B}=0$ (b) Coplanar

(c) Both null vectors (d) $\vec{A}\cdot\vec{B}=0$

7. $\vec{A}\times\vec{A}$ is :

(a) 0 (b) 1

(c) -1 (d) None of these

8. Which of the following is not a vector quantity ?

(a) Pressure (b) Force

(c) Momentum (d) Velocity

9. The vectors are equal if :

(a) They have equal magnitudes

(b) They have same directions

(c) They have equal magnitudes and same direction

(d) They have equal magnitudes and opposite directions

10. Dot product of two vectors give :

(a) Scalars quantity (b) Vector quantity

(c) Null vector (d) None of these

11. If l, m, n are direction cosines of a vector, then :

(a) $l^2+m^2+n^2=0$ (b) $l^2+m^2+n^2=1$

(c) $l^2+m^2+n^2>1$ (d) $l^2+m^2+n^2<1$

12. Volume of parallelopiped formed by $\vec{A}, \vec{B}$ and $\vec{C}$ is :

(a) $\vec{A}\times(\vec{B}\cdot\vec{C})$ (b) $\vec{A}(\vec{B}\times\vec{C})$

(c) $\vec{A}\cdot(\vec{B}\cdot\vec{C})$ (d) $\vec{A}\cdot(\vec{B}\times\vec{C})$

13. A vector $\vec{A}$ points vertically upward and $\vec{B}$ points towards north. Then $\vec{A}\times\vec{B}$ will be :

(a) Along west (b) Along east

(c) Zero (d) Vertically downward

14. The magnitude of the vector product of two vectors $\vec{A}$ and $\vec{B}$ cannot be:

(a) Equal to AB (b) Less than AB

(c) Zero (d) Greater than AB

15. Which of the following depends on the orientation of the axes?

(a) A scalar (b) A vector

(c) Component of a vector (d) Magnitude of vector

16. The maximum value of angle between two vectors is :

(a) 0 (b) $\pi/2$

(c) π (d) 2π

17. The value of $\vec{A}\cdot(\vec{B}\times\vec{C})$ for coplanar vectors is :

(a) A (b) B

(c) 0 (d) 1

18. Two vectors $\vec{A}$ and $\vec{B}$ are equal to each other if :

(a) $\vec{A}\times\vec{B}=0$ (b) $\vec{A}\cdot\vec{B}=0$

(c) $\vec{A}\cdot\vec{B}=1$ (d) $\vec{A}\times\vec{B}=1$

19. The cross product of two vectors gives :

(a) A scalar quantity (b) Vector quantity

(c) A constant (d) None of these

20. Two vectors $\vec{A}$ and $\vec{B}$ are collinear if :

(a) $\vec{A}\cdot\vec{B}=0$ (b) $\vec{A}\times\vec{B}=0$

(c) $\vec{A}\cdot\vec{B}=1$ (d) $\vec{A}\times\vec{B}=1$

21. Three vectors $\vec{A}, \vec{B}$ and $\vec{C}$ will form a vector triangle if :

(a) $\vec{A}+\vec{B}>\vec{C}$ (b) $\vec{A}+\vec{B}<\vec{C}$

(c) $\vec{A}+\vec{B}=\vec{C}$ (d) $\vec{A}+\vec{B}+\vec{C}=0$

22. The value of $\hat{A}\cdot\hat{A}$ is :

(a) A (b) A^2

(c) 0 (d) 1

SHORT ANSWER TYPE QUESTIONS

1. A vector has zero magnitude. What can you say about its direction?

Ans: A zero vector is said to have no direction, or thought of in another way, it has all directions.

2. If $\vec{A}\times\vec{B}=0$, is it necessary that $\vec{A}$ & $\vec{B}$ must be parallel ?

Ans: The given statement is true only if $\vec{A},\vec{B}\neq 0.\vec{A}\times\vec{B}=0$ can also be zero when the lines are antiparallel.

3. If $\hat{i}$ is unit vector along east and $\hat{j}$ along south, what is the unit vector along south east?

Ans: The unit vector along an arbitrary direction is given as

$$\hat{\mathrm{A}}=\frac{\vec{A}}{|A|}$$

The vector along south-east direction is a sum of vector along south and east, thus

$$\vec{\mathrm{A}}=\hat{i}+\hat{j}$$

Hence, $|\mathrm{A}|=\sqrt{2}$, so that $\hat{\mathrm{A}}=\dfrac{\hat{i}+\hat{j}}{\sqrt{2}}$ is the unit vector along south-east direction.

4. Is it possible for a vector to have a magnitude equal to one of its components?

Ans: No, a vector cannot have a magnitude equal to one of its components until all the components are non-zero and real.

5. State the condition for three vectors $\vec{A},\vec{B}$ and $\vec{C}$ to be coplanar.

Ans: Three non-zero vectors are said to be coplanar if they lie in the same plane. Thus $\vec{A},\vec{B}$ and $\vec{C}$ be three non-zero vectors if $\vec{C}$ can be written as a linear combination of $\vec{A}$ and $\vec{B}$, such that

$$\vec{\mathrm{C}}=\alpha\vec{\mathrm{A}}+\beta\vec{\mathrm{B}}$$

and if the scalar triple product of these vectors is zero i.e., if $\vec{A}\cdot\left(\vec{B}\times\vec{C}\right)=0$

6. If $|\vec{A}+\vec{B}|=|\vec{A}-\vec{B}|$, what is the angle between $\vec{A}$ and $\vec{B}$?

Ans: Given, $|\vec{A}+\vec{B}|=|\vec{A}-\vec{B}|$

Squaring both sides, one gets

$$|\vec{A}+\vec{B}|^2=|\vec{A}-\vec{B}|^2$$

or $$\vec{A}\cdot\vec{A}+\vec{B}\cdot\vec{B}+2\vec{A}\cdot\vec{B}=\vec{A}\cdot\vec{A}+\vec{B}\cdot\vec{B}-2\vec{A}\cdot\vec{B}$$

$$2\vec{A}\cdot\vec{B}=-2\vec{A}\cdot\vec{B}$$

or $$\vec{A}\cdot\vec{B}=0$$

$\vec{A}$ and $\vec{B}$ are perpendicular to each other as dot product of two non-zero vectors is zero for two perpendicular vectors.

7. A quantity possesses both magnitudes and direction. It is necessarily a vector? Explain.

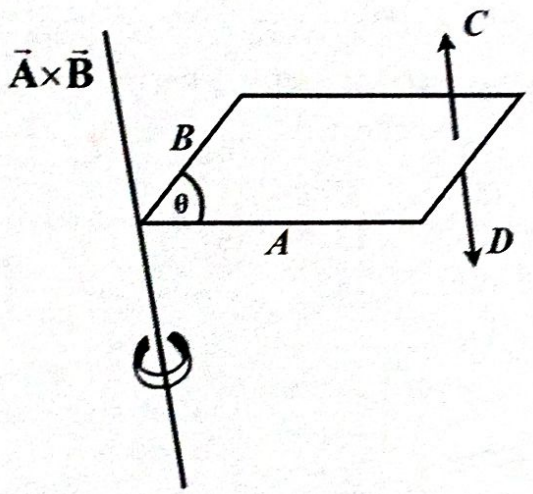
Ans: No, it is not necessary that a quantity possessing magnitude and direction are vectors. The electric current has both direction and magnitude, but it is not a vector. A vector should also have definite location in space.

8. What is the difference between the number 0 and the vector $\vec{0}$?

Ans: The number 0 is a scalar quantity as it has magnitude only, whereas the vector $\vec{0}$ has both direction and magnitude.

9. Give the vector diagram representation of $\vec{A}\times\vec{B}=\vec{C}$ and $\vec{B}\times\vec{A}=\vec{D}$. Name a vector physical quantity which is the product of two vectors.

Ans: The diagrammatic representation of two given products can be shown in a single diagram is shown below



Fig

The vector quantities that can be defined in terms of cross product of two vectors a..

(a) Torque

(b) Angular momentum

10. How do you define the direction of angular momentum? Why is the Higgs particle termed as Higgs scalar not Higgs vector?

Ans: The direction of angular momentum is perpendicular to the plane of rotation. Using the right-hand rule, the direction of angular momentum is defined as the direction in which the thumb of your right-hand points when curl of fingers represents the direction of rotation. The Higgs field is scalar because it does not act in a specific direction.

11. Consider two vectors $\vec{A}=6\hat{i}+4\hat{j}+3\hat{k}$ and $\vec{B}=2\hat{i}-3\hat{j}-3\hat{k}$. Evaluate $\vec{A}\cdot\vec{B}$ and obtain the cosine of the angle between the two.

Ans: $\vec{A}\cdot\vec{B}=\left(6\hat{i}+4\hat{j}+3\hat{k}\right)\cdot\left(2\hat{i}-3\hat{j}-3\hat{k}\right)=12-12-9=-9$

The cosine between the two vectors is given as

$$\cos\theta = \frac{\vec{A}\cdot\vec{B}}{|\vec{A}||\vec{B}|} = \frac{-9}{\sqrt{36+16+9}\sqrt{4+9+9}} = \frac{-9}{\sqrt{61}\sqrt{22}}$$

12. The coordinates of the two points P and Q are $(3,4,-6)$ and $(1,-2,3)$. Find $\overrightarrow{PQ}$.

Ans: $\overrightarrow{PQ} = Q - P = \hat{i} - 2\hat{j} + 3\hat{k} - 3\hat{i} - 4\hat{j} + 6\hat{k} = -2\hat{i} - 6\hat{j} + 9\hat{k}$

13. What is the geometrical interpretation of the vector product of two vectors?

Ans: $\vec{A}\times\vec{B}$ represents in magnitude the area of the parallelogram of which $\vec{A}$ and $\vec{B}$ are the sides drawn from a common origin (Shown in Fig. 4.1), and its direction is the normal to the plane of the parallelogram on the side on which the rotation from $\vec{A}$ to $\vec{B}$ appears counter clockwise.

14. Define scalar field and vector field in a region R in space.

Ans: A scalar field is an assignment of a scalar to each point in region in the space. e.g., the temperature at a point on the earth is a scalar field. A vector field is an assignment of a vector to each point in a region in the space. e.g., the velocity field of a moving fluid is a vector field as it associates a velocity vector to each point in the fluid.

MULTIPLE CHOICE QUESTIONS ANSWERS

1	(a)	**2**	(d)	**3**	(b)	**4**	(c)	**5**	(a)
6	(a)	**7**	(a)	**8**	(a)	**9**	(c)	**10**	(a)
11	(b)	**12**	(d)	**13**	(a)	**14**	(d)	**15**	(c)
16	(c)	**17**	(c)	**18**	(a)	**19**	(b)	**20**	(b)
21	(c)	**22**	(b)						

the prouduct of component of velocity of fluid along the direction of unit normal drawn outward. Hence $\vec{V}.\hat{n}\,ds$ represents the amount of fluid i.e., mass of fluid flowing out per unit time through element of surface ds. Now $\int_S \vec{F}.\hat{n}\,ds$ is the total mass of fluid flowing out of the surface S per unit time. Also div $\vec{V}$ gives the amount or mass of fluid passing through the point per unit time per unit volume. Then $\iiint_V \vec{\nabla}.\vec{F}\,dV$ gives the mass of fluid supplied by the sources inside per unit time.

Then from Gauss divergence theorem $\iint_S \vec{V}.\hat{n}\,ds = \iiint_V \vec{\nabla}.\vec{v}\,dV$ can be interpreted as the volume of the fluid diverging from a closed surface per unit time is equal to volume of fluid supplied from within volume per unit time.

3. Green's Identities :

(a) First Identity : If f and g are scalar point functions having continuous derivatives of second order then, $\iiint_V \left(f\nabla^2 g + \vec{\nabla}f.\vec{\nabla}g\right)dV = \iint_S \left(f\vec{\nabla}g\right).\vec{ds}$

Proof : From Gauss Divergence Theorem $\iiint_V \vec{\nabla}.\vec{F}\,dV = \iint_V \vec{F}.\vec{ds}$ .....(1)

Let $\vec{F} = f\vec{\nabla}g$

$\therefore$ Equation (1) gives $\iiint_V \vec{\nabla}.\left(f\vec{\nabla}g\right)dV = \iint_S \left(f\vec{\nabla}g\right).\vec{ds}$

or $\iiint_V \left[f\left(\vec{\nabla}.\vec{\nabla}g\right) + \vec{\nabla}f.\vec{\nabla}g\right]dV = \iint_S \left(f\vec{\nabla}g\right).\vec{ds}$

$\left\{\text{From the vector Identity } \vec{\nabla}.\left(\phi\vec{A}\right) = \phi\vec{\nabla}.\vec{A} + \vec{\nabla}\phi.\vec{A}\right\}$

or $\iiint_V \left[f\nabla^2 g + \vec{\nabla}f.\vec{\nabla}g\right]dV = \iint_S \left(f\vec{\nabla}g\right).\vec{ds}$ .....(2)

Note : We can write $\left(f\vec{\nabla}g\right).\vec{ds} = \left(f\vec{\nabla}g\right).\hat{n}ds$

$$= f\frac{\partial g}{\partial n}\hat{n}.\hat{n}\,ds = f\frac{\partial g}{\partial n}ds$$

Then Green's 1st identity can be rewritten as,

$$\boxed{\iiint_V \left[f\nabla^2 g + \vec{\nabla}f.\vec{\nabla}g\right]dV = \iint_S f\frac{\partial g}{\partial n}ds}$$

(b) Second Identity : $\iiint_V (f\nabla^2 g - g\nabla^2 f)dV = \iint_S \left(f\vec{\nabla} g - g\vec{\nabla} f \right).\vec{ds}$

From Green's 1st identity we have,

$$\iiint_V \left(f\nabla^2 g + \vec{\nabla} f.\vec{\nabla} g \right) dV = \iint_S \left(f\vec{\nabla} g \right).\vec{ds}$$

Interchanging f and g in equation (1) we get,

$$\iiint_V \left(g\nabla^2 f + \vec{\nabla} g.\vec{\nabla} f \right) dV = \iint_S \left(g\vec{\nabla} f \right).\vec{ds}$$

Now subtracting equation (2) from equation (1) we get,

$$\iiint_V \left[f\nabla^2 g + \vec{\nabla} f.\vec{\nabla} g - g\nabla^2 f - \vec{\nabla} f.\vec{\nabla} g \right] dV = \iint_S \left(f\vec{\nabla} g - g\vec{\nabla} f \right).\vec{ds}$$

or $\iiint_V \left[f\nabla^2 g - g\nabla^2 f \right] dV = \iint_S \left[f\vec{\nabla} g - g\vec{\nabla} f \right].\vec{ds}$ (Proved)

The other form of Green's 2nd identity can be expressed as

$$\iiint_V \left[f\nabla^2 g - g\nabla^2 f \right] dV = \iint_S \left[f\vec{\nabla} g - g\vec{\nabla} f \right].\hat{n}\,\vec{ds}$$

or $$\iiint_V \left[f\nabla^2 g - g\nabla^2 f \right] dV = \iint_S \left[f\frac{\partial g}{\partial n}\hat{n} - g\frac{\partial f}{\partial n}\hat{n} \right].\hat{n}\,\vec{ds}$$

or $$\boxed{\iiint_V \left[f\nabla^2 g - g\nabla^2 f \right] dV = \iint_S \left[f\frac{\partial g}{\partial n} - g\frac{\partial f}{\partial n} \right]\vec{ds}}.$$